James Madison University
Information Technology (IT) Program
IT 333 Advanced Networking for IT
Fall 2026

Lab 2 - Network Devices, Functions, and services

SecGrp 3-11

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13:03:36

Author(s):
Matthew Henriques
Ahmed Elgamal

Submitted to:
Dr. Emil Salib



Honor Pledge: I have neither given nor received help on his lab that violates the spirit of the JMU Honor Code.

Matthew Henriques	Ahmed Elgamal
Matthew Henriques	*Ahmed Elgamal*
Signature	*Signature*
9282026	*9282026*
Date	*Date*

Execution: Here is my percentage of executing the the hands-on portion of the lab. It is expected that each partner executes 100%.

Matthew Henriques (100%)	Ahmed Elgamal (100%)
Matthew Henriques	*Ahmed Elgamal*
Signature	*Signature*
9282026	*9282026*
Date	*Date*

Contributions: Here is my Lab Report Contribution Percentages for this lab. It is expected that the sum of the contributions of the group members is 100%.

Matthew Henriques (50%)	Ahmed Elgamal (50%)
Matthew Henriques	*Ahmed Elgamal*
Signature	*Signature*
9282026	*9282026*
Date	*Date*

Contributions Agreement and Confirmation: In the case of the contribution percentage is not the same for all students, has each of you agreed on and sent the instructor a confirmation email of the percentage above?

Matthew Henriques (Yes)	Ahmed Elgamal (Yes)
Matthew Henriques	*Ahmed Elgamal*
Signature	*Signature*
9282026	*9282026*
Date	*Date*

Rubric Check: I have read and complied with every item listed in this lab's rubric as posted on Canvas.

Matthew Henriques (YES)	Ahmed Elgamal (YES)
Matthew Henriques	*Ahmed Elgamal*
Signature	*Signature*
9/28/2026	*9/28/2026*
Date	*Date*

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65          /L2E1S2T3-7.png
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67          /L2E1S2T3-5.png
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74          /L2E1S2T4-11.png
75          /L2E1S2T4-8.png
76          /L2E1S2T4-7.png
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```

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    L2E1S3
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107          /L2E1S3T3-10.png
    L2E1S3/L2E1S3T4
108          /L2E1S3T4-3.png
109          /L2E1S3T4-1.png
110          /L2E1S3T4-2.png
    L2E1S3/L2E1S3T5
111          /L2E1S3T5-7.png
112          /L2E1S3T5-1.png
113          /L2E1S3T5-8.png
114          /L2E1S3T5-4.png
115          /L2E1S3T5-2.png
116          /L2E1S3T5-5.png
117          /L2E1S3T5-3.png
118          /L2E1S3T5-6.png
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120          /L2E1S3T6-3.png
121          /L2E1S3T6-1.png
122          /L2E1S3T6-4.png
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1 Introduction

Value and Impact:

This lab was valuable to us because it showed us how many essential network services and protocols work together to provide the normal communication that many of us take for granted. According to [1], ARP, DNS, and DHCP are all used extensively behind the scenes, and understanding them is important for troubleshooting connectivity problems and configuring networks properly. We really enjoyed this lab due to the use of Wireshark and the ability to analyze the actual packets and handshakes involved in these protocols. It is also very valuable to us since we both want to go into Cyber Security and it is essential to understand how these protocols work on the network since you can't secure networks without this fundamental knowledge.

Learning Objectives:

The learning objectives of Step 1 are to understand how ARP works and how IPv4 addresses are matched to EUI-48 addresses on a local network. We also learned how to view ARP packets in Wireshark and how arping, ICMP, and ARP cache are all related.

The learning objectives of Step 2 are to learn how different types of name resolutions work such as mDNS, normal DNS resolution, and iterative DNS lookups. We also learned how changing the DNS Server can change where DNS requests are sent.

The learning objectives of Step 3 (if any) are to understand how DHCP gives network settings to clients. We also learned how a DHCP relay helps clients communicate with a DHCP server outside of its subnet, and how DHCP packets change when they pass through the relay.

2 Lab Exercises: Network Diagrams

2.1 Exercise 1 - What are network services?

2.1.1 Step 1: ARP (Address and Resolution Protocol

Figure 1 represents the network arrangement and information for Step 1 tasks.

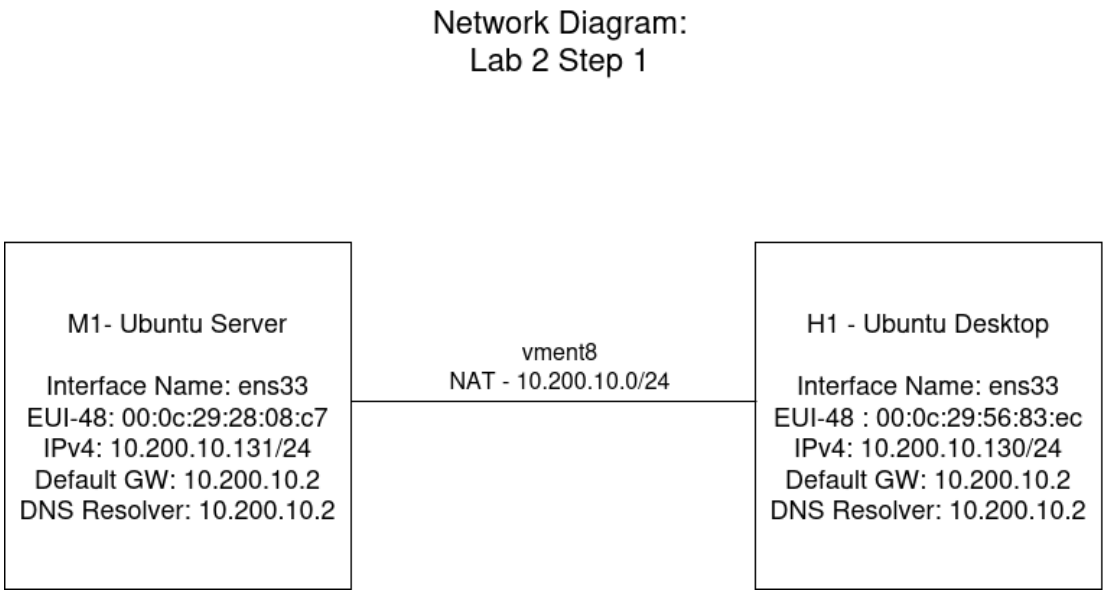


Figure 1: Step 1 Network Diagram

2.1.2 Step 2: DNS (Domain Name System) Services & Protocol

Figure 2 represents the network arrangement and information for Step 2 tasks.

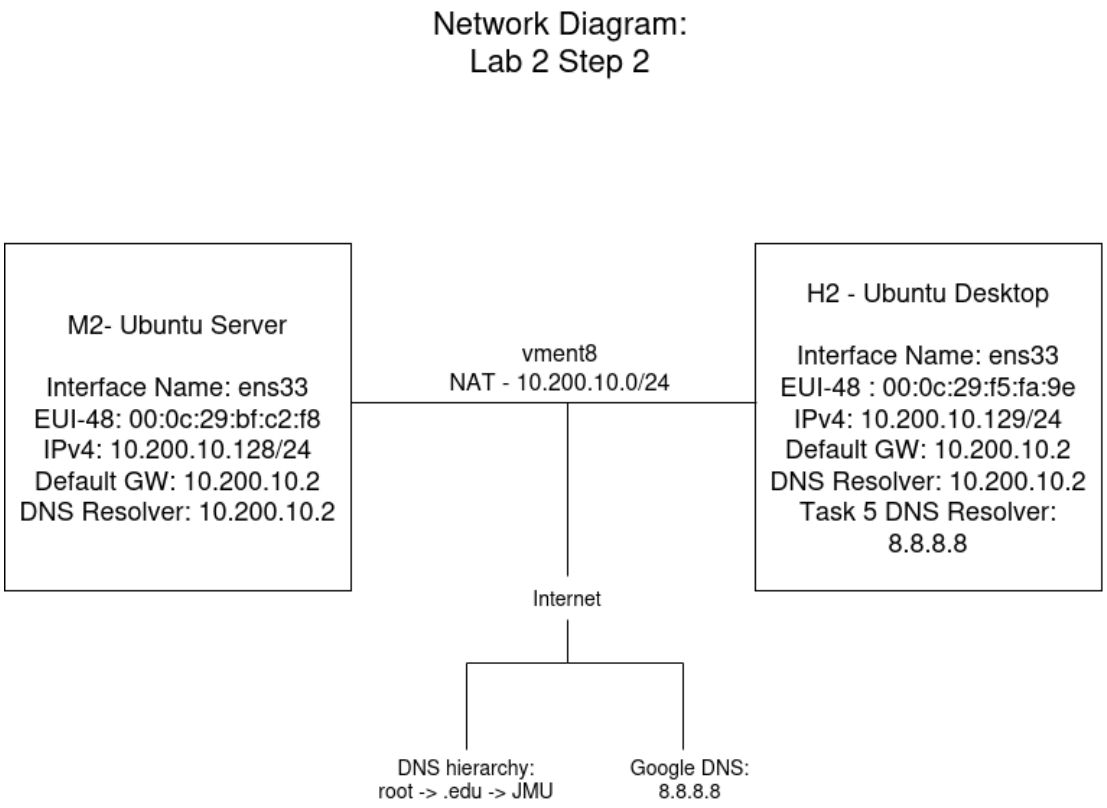


Figure 2: Step 2 Network Diagram

2.1.3 Step 3: DHCP Server and DHCP Relay for multiple subnets environment

Figure 3 represents the network arrangement and information for Step 3 tasks.

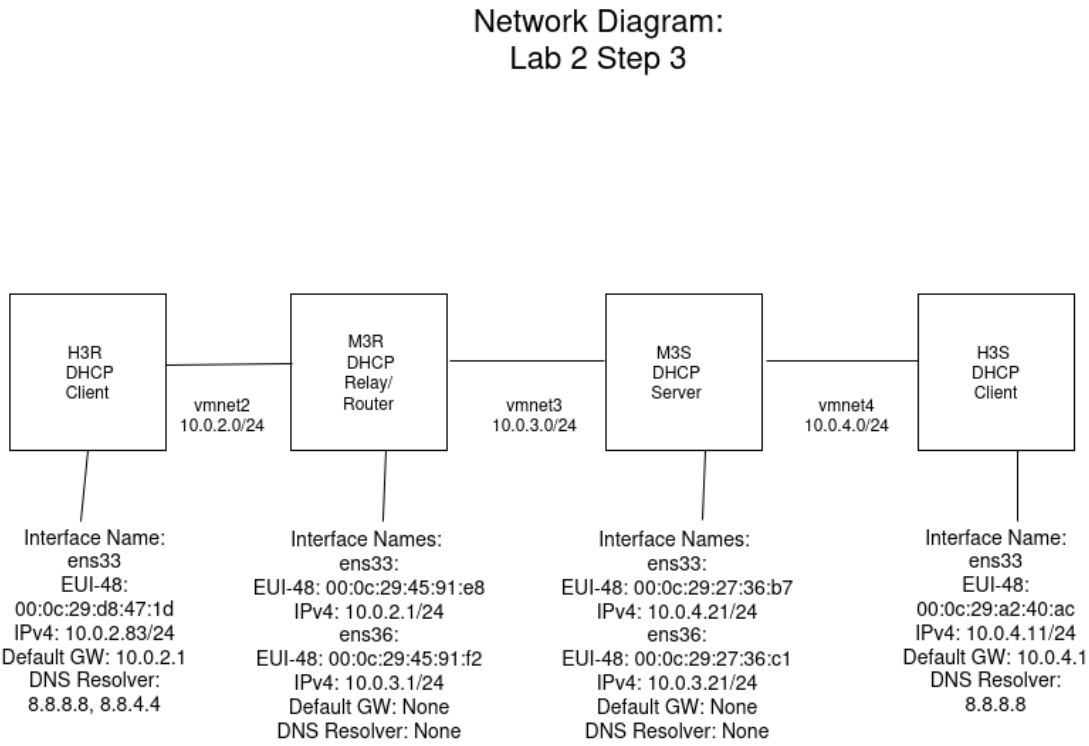


Figure 3: Step 3 Network Diagram

3 Lab Exercises : Results & Analysis

3.1 Exercise 1 - What are Network Services?

3.1.1 Step 1: ARP (Address Resolution Protocol)

Figure 1 represents the network we used to perform all or some of the tasks in Step 1.

SubA: Introduction:

T1: Done. See the evidence (screenshots in images/L2E1S1/L2E1S1T1 subfolder and attachments, if any, in atts/L2E1S1/L2E1S1T1 subfolder.)

Q1: The question is

(a) State whether you are using VMware WS or VMware Fusion and provide the release level.

(b) What is the general guideline of default gateway IPv4 address selection that VMware WS has broken in the case of VMware WS and has recently corrected on VMware Fusion?

(c) Provide evidence of the IPv4 address assigned by VMware to the default gateway. Provide evidence of the IPv4 addresses that are assigned to the Host (physical machine) virtual interface for vmnet1 (host-only) and vmnet8 (NAT).

(d) Explain why VMware WS has made that choice .

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A1: The answer is

(a) We used VMware Workstation Pro version 26H1 for this part of the lab. Figure 4 shows the VMware Workstation release level that was used.

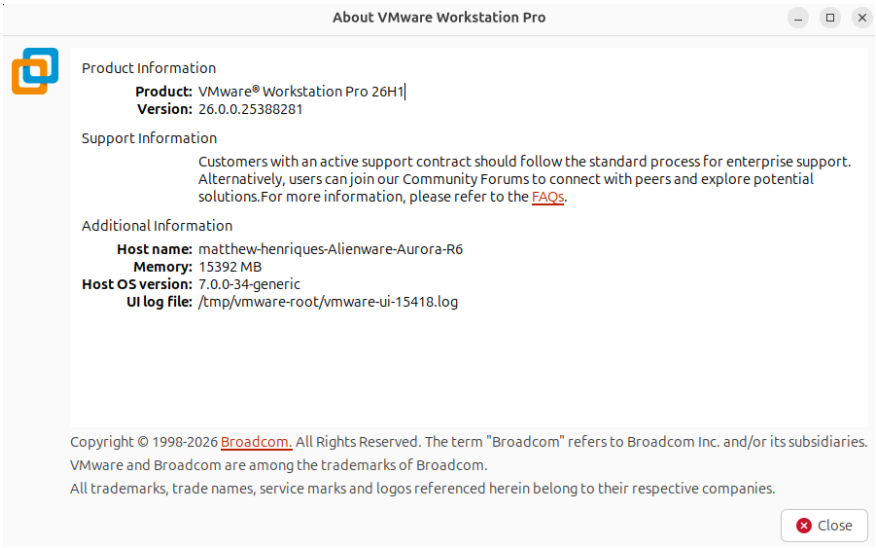


Figure 4: VMware Workstation Pro Version

(b) The general guideline is to use the first usable IPv4 address in the subnet as the default gateway. In this VMware NAT network, the first usable address, 10.200.10.1, was assigned to the host's vmnet8 virtual interface instead, while the default gateway was assigned 10.200.10.2. This lab notes that VMware Workstation breaks this normal guideline, while the newer versions of VMware Fusion have corrected it. Figure 5 shows the host vmnet8 address, while Figure 6 shows 10.200.10.2 as the M1's default gateway.

```
matthew-henriques@matthew-henriques-Alienware-Aurora-R6:~$ ip -4 addr show vmnet1
5: vmnet1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    altname enx005056c00001
    inet 10.100.10.1/24 brd 10.100.10.255 scope global vmnet1
       valid_lft forever preferred_lft forever
matthew-henriques@matthew-henriques-Alienware-Aurora-R6:~$ ip -4 addr show vmnet8
6: vmnet8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    altname enx005056c00008
    inet 10.200.10.1/24 brd 10.200.10.255 scope global vmnet8
       valid_lft forever preferred_lft forever
matthew-henriques@matthew-henriques-Alienware-Aurora-R6:~$
```

Figure 5: Host virtual interfaces showing the IPv4 address assigned to vmnet8.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip route show
default via 10.200.10.2 dev ens33 proto dhcp src 10.200.10.131 metric 100
10.200.10.0/24 dev ens33 proto kernel scope link src 10.200.10.131 metric 100
10.200.10.2 dev ens33 proto dhcp scope link src 10.200.10.131 metric 100
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 6: M1 routing table showing 10.200.10.2 as the vment 8 default gateway.

(c) VMware assigned 10.200.10.2 as the default gateway for the vm-

net8 NAT subnet. The physical host's vmnet1 virutal interface used 10.100.10.1, while the vmnet8 virtual interface used 10.200.10.1. Figures 7 8 9 10 show the VMware network configuration and the IPv4 addresses assigned to the host virtual interfaces.

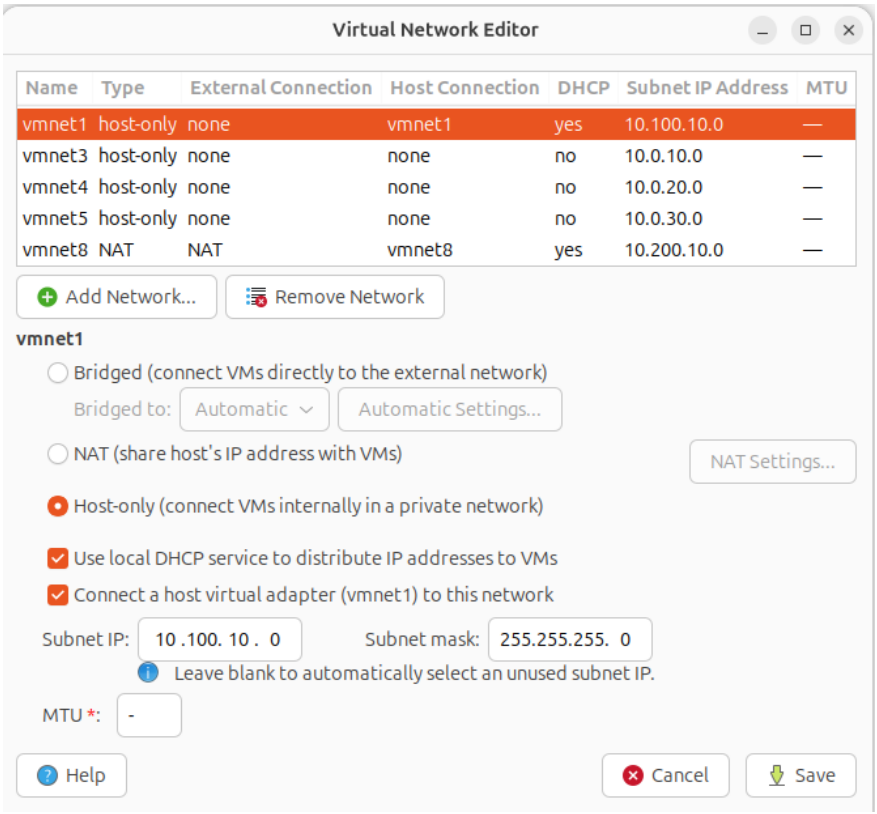


Figure 7: VMware vmnet1 network configuration.

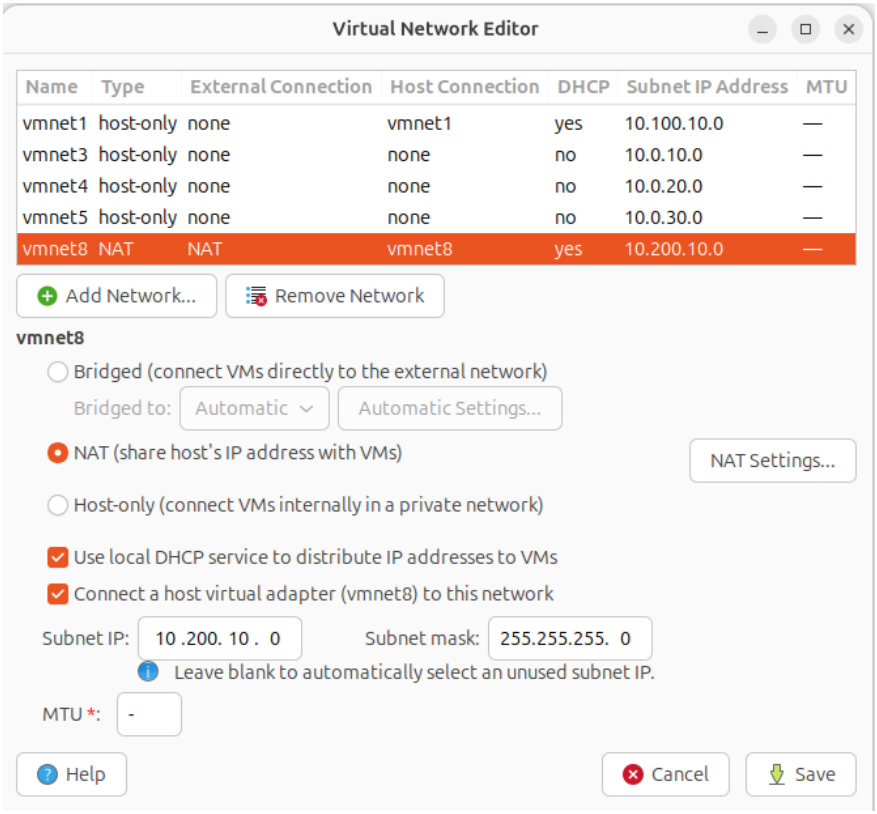


Figure 8: VMware vmnet8 network configuration.

```
matthew-henriques@matthew-henriques-Alienware-Aurora-R6:~$ ip -4 addr show vmnet1
5: vmnet1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    altname enx005056c00001
    inet 10.100.10.1/24 brd 10.100.10.255 scope global vmnet1
       valid_lft forever preferred_lft forever
matthew-henriques@matthew-henriques-Alienware-Aurora-R6:~$ ip -4 addr show vmnet8
6: vmnet8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    altname enx005056c00008
    inet 10.200.10.1/24 brd 10.200.10.255 scope global vmnet8
       valid_lft forever preferred_lft forever
matthew-henriques@matthew-henriques-Alienware-Aurora-R6:~$
```

Figure 9: Physical host virtual interface addresses for vmnet1 and vmnet8.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip route show
default via 10.200.10.2 dev ens33 proto dhcp src 10.200.10.131 metric 100
10.200.10.0/24 dev ens33 proto kernel scope link src 10.200.10.131 metric 100
10.200.10.2 dev ens33 proto dhcp scope link src 10.200.10.131 metric 100
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 10: M1 routing table showing 10.200.10.2 as the vmnet8 default gateway.

(d) According to Broadcom [2], Workstation uses separate virtual components for the physical host's virtual adapter and the NAT gateway. In our configuration, according to Figures 11 and 12 the host virtual adapter used 10.200.10.1, while the VMware NAT gateway used 10.200.10.2. This allows the host itself and the NAT service to have seperate addresses on the same virtual subnet.

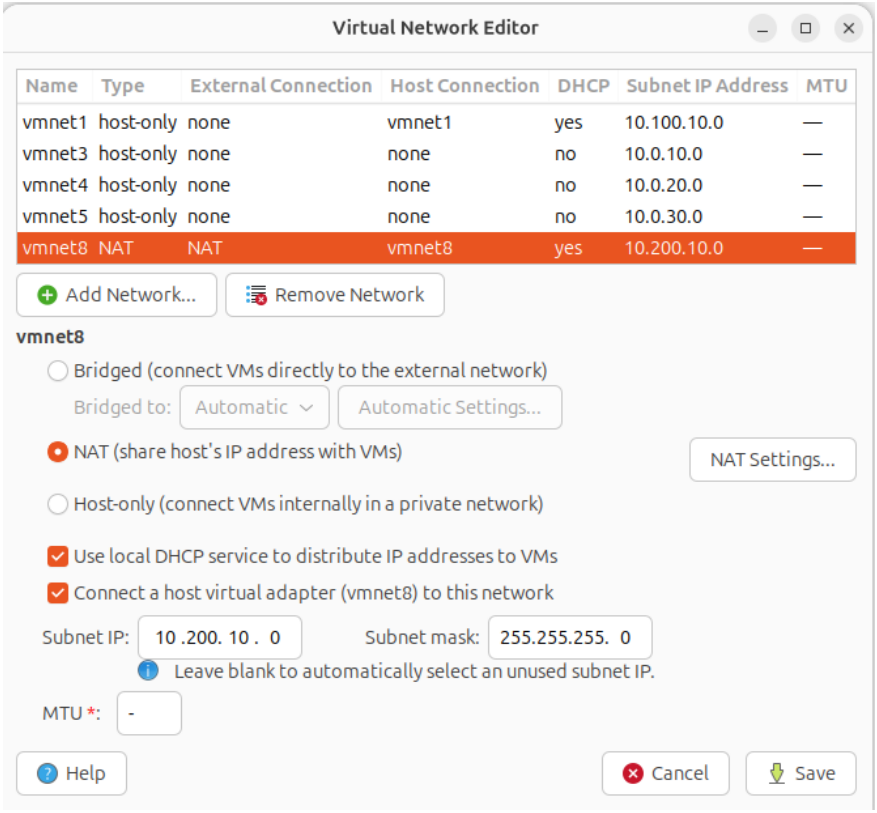


Figure 11: VMware vmnet8 network configuration.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip route show
default via 10.200.10.2 dev ens33 proto dhcp src 10.200.10.131 metric 100
10.200.10.0/24 dev ens33 proto kernel scope link src 10.200.10.131 metric 100
10.200.10.2 dev ens33 proto dhcp scope link src 10.200.10.131 metric 100
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 12: M1 routing table shoing 10.200.10.2 as the vmnet8 default gateway.

T2: Done. See the evidence (screenshots in images/L2E1S1/L2E1S1T2 subfolder and attachments, if any, in atts/L2E1S1/L2E1S1T2 subfolder.)

Q2: The question is
Explain why you are asked to delete the content of the dhclient.leases file. Hint: dhcp server.
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A2: The answer is
The dhclient.leases file was cleared so M1 would not reuse information from a previously stored DHCP lease. Deleting the lease information in this file allowed the DHCP process to obtain fresh information from the DHCP server. Figure 13 shows the existing lease before it was cleared, and Figure 14 shows the empty lease file afterward.

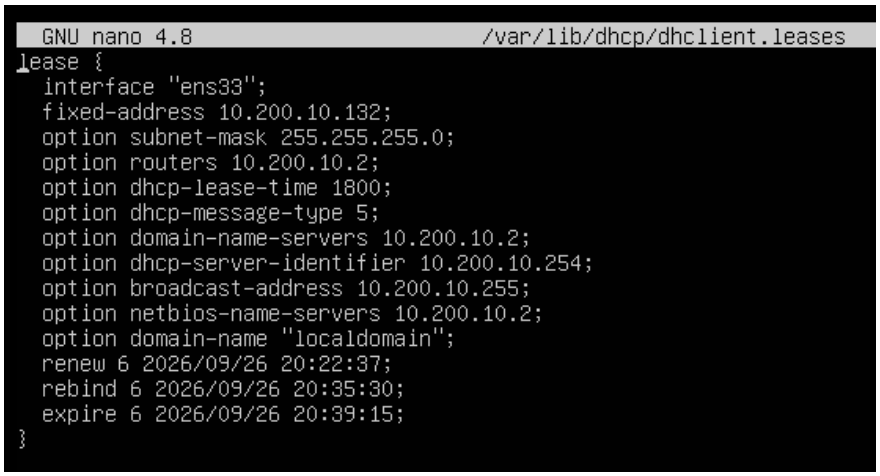


Figure 13: M1 DHCP lease file before clearing the previously stored lease information.

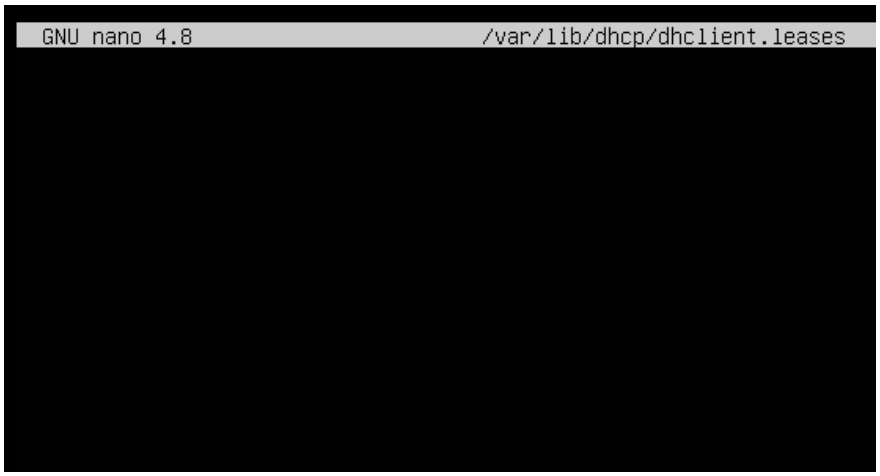


Figure 14: M1 DHCP lease file after the previous lease information was removed.

Q3: The question is
What is the default gateway IPv4 address of the NAT subnet (vmnet8) and the corresponding EUI-48 address?
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A3: The answer is
The default gateway of the vmnet8 NAT subnet was 10.200.10.2, and its corresponding EUI-48 adress was 00:50:56:f5:cc:cc. Figure 15 shows 10.200.10.2 as the M1's default gateway, while Figure 16 shows the neighbor-table entry mapping that IPv4 address to its EUI-48 address.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip route show
default via 10.200.10.2 dev ens33 proto dhcp src 10.200.10.131 metric 100
10.200.10.0/24 dev ens33 proto kernel scope link src 10.200.10.131 metric 100
10.200.10.2 dev ens33 proto dhcp scope link src 10.200.10.131 metric 100
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 15: M1 routing table showing the vmnet8 default gateway at 10.200.10.2

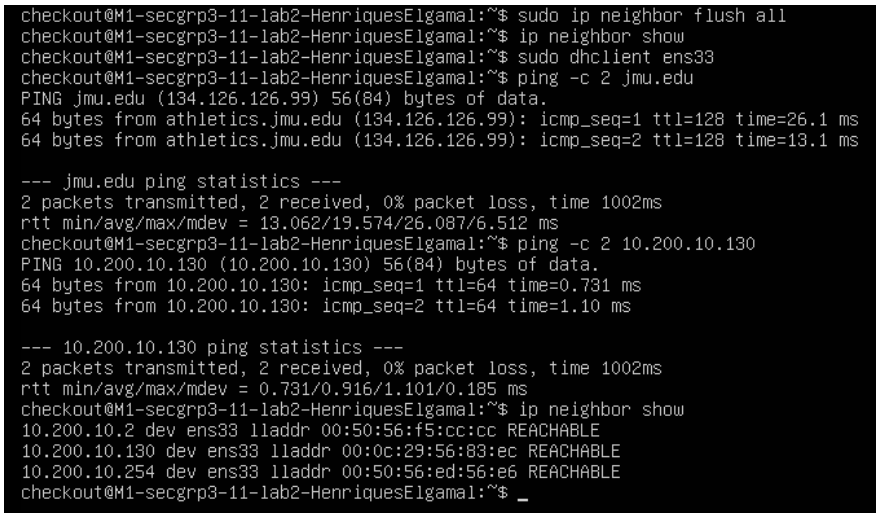


Figure 16: M1 neighbor table at the bottom showing the EUI-48 address associated with the 10.200.10.2 default gateway

Q4: The question is
What is the IPv4 address assigned by VMware to the DHCP server serving the vmnet8 virtual network? Explain and provide evidence as to how you determined that the IPv4 address is that of the vmnet8 DHCP server. Hint: lease.
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A4: The answer is
According to Figure 17, the DHCP server serving vmnet8 had the IPv4 address of 10.200.10.254 shown in M1's DHCP lease file.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ cat /var/lib/dhcp/dhclient.leases
lease {
  interface "ens33";
  fixed-address 10.200.10.131;
  option subnet-mask 255.255.255.0;
  option routers 10.200.10.2;
  option dhcp-lease-time 1800;
  option dhcp-message-type 5;
  option domain-name-servers 10.200.10.2;
  option dhcp-server-identifier 10.200.10.254;
  option broadcast-address 10.200.10.255;
  option netbios-name-servers 10.200.10.2;
  option domain-name "localdomain";
  renew 6 2026/09/26 20:38:14;
  rebind 6 2026/09/26 20:50:09;
  expire 6 2026/09/26 20:53:54;
}
lease {
  interface "ens33";
  fixed-address 10.200.10.132;
  option subnet-mask 255.255.255.0;
  option routers 10.200.10.2;
  option dhcp-lease-time 1800;
  option dhcp-message-type 5;
  option domain-name-servers 10.200.10.2;
  option dhcp-server-identifier 10.200.10.254;
  option broadcast-address 10.200.10.255;
  option netbios-name-servers 10.200.10.2;
  option domain-name "localdomain";
  renew 6 2026/09/26 20:36:33;
  rebind 6 2026/09/26 20:50:11;
  expire 6 2026/09/26 20:53:56;
}
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 17: M1 DHCP lease showing 10.200.10.254 as the vmnet8 DHCP server identifier.

Q5: The question is
Explain the differences among the following arp record's states: REACHABLE, STALE, DELAY, and FAILED. Hint: ip neighor or ip neigh show of arp/neighbor records.
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A5: The answer is
According to ChatGPT[3], REACHABLE means the device was contacted recently successfully so the IP to MAC mapping is still valid. STALE means the mapping is known but has not been confirmed recently so it can still be used but Linux may verify it again. DELAY means Linux is waiting briefly before checking a stale neighbor is still reachable. FAILED means that Linux attempted the ARP resolution but no response was received. Figure 18 shows neighbor entries in the REACHABLE state, while Figures 19 and 20 show additional neighbor states that occured during this lab.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ sudo ip neighbor flush all
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ sudo dhclient ens33
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ping -c 2 jmu.edu
PING jmu.edu (134.126.126.99) 56(84) bytes of data.
64 bytes from athletics.jmu.edu (134.126.126.99): icmp_seq=1 ttl=128 time=26.1 ms
64 bytes from athletics.jmu.edu (134.126.126.99): icmp_seq=2 ttl=128 time=13.1 ms

--- jmu.edu ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 13.062/19.574/26.087/6.512 ms
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ping -c 2 10.200.10.130
PING 10.200.10.130 (10.200.10.130) 56(84) bytes of data.
64 bytes from 10.200.10.130: icmp_seq=1 ttl=64 time=0.731 ms
64 bytes from 10.200.10.130: icmp_seq=2 ttl=64 time=1.10 ms

--- 10.200.10.130 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.731/0.916/1.101/0.185 ms
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
10.200.10.2 dev ens33 lladdr 00:50:56:f5:cc:cc REACHABLE
10.200.10.130 dev ens33 lladdr 00:0c:29:56:83:ec REACHABLE
10.200.10.254 dev ens33 lladdr 00:50:56:ed:56:e6 REACHABLE
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 18: M1 neighbor table showing active entries

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
10.200.10.2 dev ens33 lladdr 00:50:56:f5:cc:cc DELAY
10.200.10.130 dev ens33 lladdr 00:0c:29:56:83:ec REACHABLE
10.200.10.254 dev ens33 lladdr 00:50:56:ed:56:e6 REACHABLE
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 19: M1 neighbor table showing the DELAY state

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
10.200.10.2 dev ens33 lladdr 00:50:56:f5:cc:cc STALE
10.200.10.254 dev ens33 lladdr 00:50:56:ed:56:e6 STALE
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 20: M1 neighbor table showing the STALE state

T3: Done. See the evidence (screenshots in images/L2E1S1/L2E1S1T3 subfolder and attachments, if any, in atts/L2E1S1/L2E1S1T3 subfolder.)

Q6: The question is

(a) Explain why the arp exchange took place on sending the ping command from M1 to H1.

(b) Would that happen if arp cache has an active (Reachable) arp record for H1. Provide Wireshark evidence in support of your claim. Hint: ping

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific de-

scription of the relevant information. Provide citations to external sources.

A6: The answer is

(a) The ARP exchange occured because M1 knew H1's IPv4 address, 10.200.10.130, but did not yet have H1's EUI-48 address in its neighbor table. Before M1 could send the Ethernet frame carrying the ICMP Echo Request, it first needs H1's Ethernet address. M1 therefore sends an ARP request. Figure 21 shows the four packets in order: ARP request, ARP reply, ICMP Echo Request, and ICMP Echo Reply.

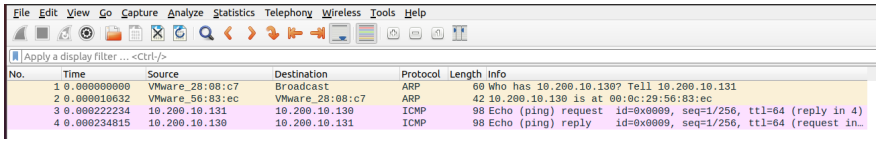


Figure 21: Wireshark capture showing the ARP request and reply followed byu the ICMP Echo Request and Echo Reply between M1 and H1

(b) If M1 already has an active REACHABLE neighbor entry for H1, another ARP exchange is not normally necessary because M1 already knows H1's EUI-48 address. Figure 22 shows H1 at 10.200.10.130 with EUI-48 address of 00:0c:29:56:83:ec in the REACHABLE state.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
10.200.10.2 dev ens33 lladdr 00:50:56:f5:cc:cc DELAY
10.200.10.130 dev ens33 lladdr 00:0c:29:56:83:ec REACHABLE
10.200.10.254 dev ens33 lladdr 00:50:56:ed:56:e6 REACHABLE
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 22: H1 at 10.200.10.130 with EUI-48 address 00:0c:29:56:83:ec in the REACHABLE state

T4: Done. See the evidence (screenshots in images/L2E1S1/L2E1S1T4 subfolder and attachments, if any, in atts/L2E1S1/L2E1S1T4 subfolder.)

Q7: The question is

(a) Provide the information carried in the payload of the ARP request.

(b) Verify the ARP payload information using information obtained from the like of "ip a", etc.

(c) Is the ARP payload information part of the network layer header? Hint: tricky!

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A7: The answer is

(a) The ARP request contained the following information: Sender IPv4 address: 10.200.10.131, Sender EUI-48 address: 00:0c:29:28:08:c7, Target EUI-48 address: 00:00:00:00:00:00, Target IPv4 address: 10.200.10.130. Figure 23 shows these fields in the ARP request

```
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: VMware_28:08:c7 (00:0c:29:28:08:c7)
    Sender IP address: 10.200.10.131
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.200.10.130
```

Figure 23: ARP request from M1 showing the sender and target IPv4 and EUI-48 fields.

(b) The sender information in the ARP request matches M1's interface information. M1's ens33 interface used IPv4 address 10.200.10.131 and EUI-48 address 00:0c:29:28:08:c7. The target IPv4 address 10.200.10.130 matches H1's ens33 address. Figures 24 and 25 used to verify the ARP payload.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:28:08:c7 brd ff:ff:ff:ff:ff:ff
    inet 10.200.10.131/24 metric 100 brd 10.200.10.255 scope global dynamic ens33
       valid_lft 1585sec preferred_lft 1585sec
    inet 10.200.10.132/24 brd 10.200.10.255 scope global secondary dynamic ens33
       valid_lft 1501sec preferred_lft 1501sec
    inet 10.200.10.133/24 brd 10.200.10.255 scope global secondary dynamic ens33
       valid_lft 1597sec preferred_lft 1597sec
    inet6 fe80::20c:29ff:fe28:8c7/64 scope link
       valid_lft forever preferred_lft forever
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 24: M1 ens33 interface information used to verify the sender fields in the ARP request.

```
checkout@H1-secgrp3-11-lab2-HenriquesElgamal:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:56:83:ec brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.200.10.130/24 brd 10.200.10.255 scope global dynamic noprefixroute e
ns33
       valid_lft 1067sec preferred_lft 1067sec
    inet6 fe80::6ff9:9cd0:c620:f799/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 25: H1 ens33 interface information used to verify the target IPv4 address in the ARP request.

(c) No, the ARP payload information is not part of the IPv4 network-layer header. According to Geek for Geeks [4], ARP has its own message format containing sender and target protocol and hardware addresses. Figure 26, shows Ethernet 2 followed directly by ARP rather than an IPv4 header.

```
▸ Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface ens33, id 0
▾ Ethernet II, Src: VMware_28:08:c7 (00:0c:29:28:08:c7), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▸ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  ▸ Source: VMware_28:08:c7 (00:0c:29:28:08:c7)
    Type: ARP (0x0806)
    Padding: 000000000000000000000000000000000000
▸ Address Resolution Protocol (request)
```

Figure 26: Wireshark showing the ARP message directy inside an Ethernet Frame without an IPv4 header.

Q8: The question is

(a) Which device (on what host) and network interface (in terms of EUI-48 address) replied to the ARP request?

(b) Is the destination EUI-48 address in the data link layer header of the ARP's reply the same as the Target EUI-48 address given as part of the data (payload) of the ARP reply?

(c) What is the sender IPv4 address of the arp reply as given in arp payload?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A8: The answer is

(a) H1 replied to the ARP request through its ens33 interface. H1's EUI-48 address was 00:0c:29:56:83:ec. Figure 27 shows this address in the ARP reply, and Figure 28 shows the same address assigned to the H1's ens33 interface.

```
▾ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: VMware_56:83:ec (00:0c:29:56:83:ec)
    Sender IP address: 10.200.10.130
    Target MAC address: VMware_28:08:c7 (00:0c:29:28:08:c7)
    Target IP address: 10.200.10.131
```

Figure 27: ARP rpely from H1 showing H1 as the sender of the response.

```
checkout@H1-secgrp3-11-lab2-HenriquesElgamal:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:56:83:ec brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.200.10.130/24 brd 10.200.10.255 scope global dynamic noprefixroute e
ns33
       valid_lft 1067sec preferred_lft 1067sec
    inet6 fe80::6ff9:9cd0:c620:f799/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 28: H1 ens33 interface information shows its assigned IPv4 address and EUI-48 address.

(b) Yes. The destination EUI-48 address in the Ethernet header and the Target EUI-48 address contained in the ARP reply were both 00:0c:29:28:08:c7, which is M1's EUI-48 address. Figure 29 shows both values in the ARP reply section.

Wireshark · Packet 2 · SecGrp3-11-L2E1S1T3-2arp-2icmp-from-M1-to-H1-on-H1-ens33.pcapng

```
▸ Frame 2: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface ens33, id 0
▾ Ethernet II, Src: VMware_56:83:ec (00:0c:29:56:83:ec), Dst: VMware_28:08:c7 (00:0c:29:28:08:c7)
  ▾ Destination: VMware_28:08:c7 (00:0c:29:28:08:c7)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  ▾ Source: VMware_56:83:ec (00:0c:29:56:83:ec)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 1]
▾ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: VMware_56:83:ec (00:0c:29:56:83:ec)
    Sender IP address: 10.200.10.130
    Target MAC address: VMware_28:08:c7 (00:0c:29:28:08:c7)
    Target IP address: 10.200.10.131
```

Figure 29: ARP reply showing that the Ethernet destination EUI-48 and ARP target EUI-48 both identify M1.

(c) The sender IPv4 address contained in the ARP reply was 10.200.10.130, which is H1's IPv4 address. Figure 30 shows this value in the ARP reply.

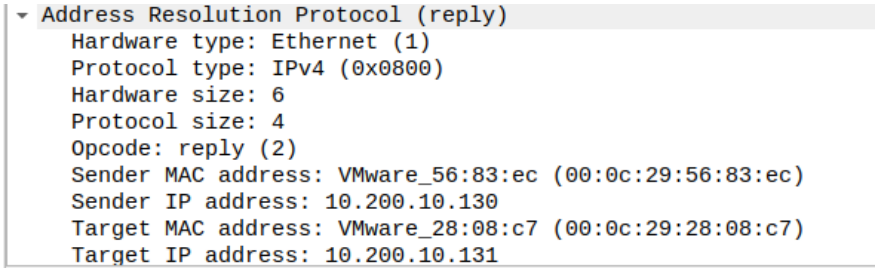


Figure 30: ARP reply showing 10.200.10.130 as the Sender IPv4 address.

T5: Done. See the evidence (screenshots in images/L2E1S1/L2E1S1T5 subfolder and attachments, if any, in atts/L2E1S1/L2E1S1T5 subfolder.)

Q9: The question is
DEEP ANALYSIS) Design, execute and document your own procedure (including commands and the responses) to verify the following statement: "using the arping command will not automatically update the arp table". You must use the commands that you have been introduced to in the above tasks. Also prove that using ICMP updates the arp cache. Hint: you should show the arp cache before and after.
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A9: The answer is
To determine whether arping automatically updates the ARP table, we first executed sudo ip neighbor flush all on M1 followed by ip neighbor show to verify there was no existing entry on for H1. Figure 31 shows the neighbor table before starting the test. We then executed arping sucessfully from M1 to H1 and Wireshark also captured two ARP requests and two ARP replies. Figures 32 and 33 show the successful arping command and the four resulting ARP packets. After arpring we checked the neighbor table again and Figure 34 shows that H1 did not appear as a normal neighbor entry therefore concluding that arping did not automatically update the ARP table in this test. After this we decided to flush the neighbor table again shown by 35 and send a normal ICMP ping to H1. After the ping completed, H1 appeared in the neighbor table at 10.200.10.130 with EUI-48 address 00:0c:29:56:83:ec in the REACHABLE state. Figure 36 shows the successful ping and neighbor table. In conclusion, our experiment showed that arping successfully exchanges ARP frames without automatically creating the normal neighbor-table entry, while a standard ICMP communication caused the neighbor cache to be updated.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ sudo ip neighbor flush all
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 31: M1 neighbor table after flushing, showing no existing H1 entry before the arping test.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ sudo arping -c 2 10.200.10.130
ARPING 10.200.10.130
60 bytes from 00:0c:29:56:83:ec (10.200.10.130): index=0 time=785.700 usec
60 bytes from 00:0c:29:56:83:ec (10.200.10.130): index=1 time=358.684 usec

--- 10.200.10.130 statistics ---
2 packets transmitted, 2 packets received,   0% unanswered (0 extra)
rtt min/avg/max/std-dev = 0.359/0.572/0.786/0.214 ms
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 32: Successful arping command from M1 to H1 with two ARP replies received.

1	0.000000000	VMware_28:08:c7	Broadcast	ARP	60	Who has 10.200.10.130? Tell
2	0.000028338	VMware_56:83:ec	VMware_28:08:c7	ARP	42	10.200.10.130 is at 00:0c:2
3	1.001145233	VMware_28:08:c7	Broadcast	ARP	60	Who has 10.200.10.130? Tell
4	1.001156641	VMware_56:83:ec	VMware_28:08:c7	ARP	42	10.200.10.130 is at 00:0c:2

Figure 33: Wireshark capture showing two ARP requests and two ARP replies generated by arping.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
10.200.10.2 dev ens33 lladdr 00:50:56:f5:cc:cc STALE
10.200.10.254 dev ens33 lladdr 00:50:56:ed:56:e6 STALE
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 34: M1 neighbor table after arping, showing that H1 was not automatically added as a normal neighbor entry.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ sudo ip neighbor flush all
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 35: M1 neighbor table after being flushed again before the ICMP test.

```
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ping -c 1 10.200.10.130
PING 10.200.10.130 (10.200.10.130) 56(84) bytes of data.
64 bytes from 10.200.10.130: icmp_seq=1 ttl=64 time=1.25 ms

--- 10.200.10.130 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.252/1.252/1.252/0.000 ms
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ ip neighbor show
10.200.10.2 dev ens33 lladdr 00:50:56:f5:cc:cc DELAY
10.200.10.130 dev ens33 lladdr 00:0c:29:56:83:ec REACHABLE
checkout@M1-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 36: Successful ICMP ping to H1 followed by H1 showing in M1's neighbor table in the REACHABLE state.

3.1.2 Step 2: DNS (Domain Name System) Services & Protocol

Figure 2 represents the network we used to perform all or some of the tasks in Step 2.

SubA: (a) /etc/hosts file Name Resolution

T1: Done. See the evidence (screenshots in images/L2E1S2/L2E1S2T1 subfolder and attachments, if any, in atts/L2E1S2/L2E1S2T1 subfolder.)

Q1: The question is

(a) What is the difference between 127.0.0.1 (localhost) and 127.0.1.1 (¡hostname¿)? Hint: See the Notes under the title Exercise for dealing with the special character in Latex.

(b) Are they considered loopback IPv4 addresses? Explain why. Hint: use ipcalc.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A1: The answer is

(a) 127.0.0.1 is associated with localhost, while 127.0.1.1 is associated with the hostname of the local machine in our /etc/hosts configuration. Both addressses however refer back to the local computer rather than another device on the network. Figure 37 shows the /etc/hosts file where 127.0.0.1 is mapped to localhost and 127.0.1.1 is mapped to the machine hostname.

```
 GNU nano 4.8                                        /etc/hosts
127.0.0.1 localhost
127.0.1.1 M2-secgrp3-11-lab2-HenriquesElgamal
10.200.10.129 H2-secgrp3-11-lab2-HenriquesElgamal

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Figure 37: H2 /etc/hosts file showing 127.0.0.1 assigned to localhost and 127.0.1.1 assigned to the H2 hostname.

(b) Yes, both addresses are loopback IPv4 addresses because they are both contained within the 127.0.0.0/8 loopback network. Our ipcalc results confirm that both 127.0.0.1 and 127.0.1.1 belong to this range. Figures 38 and 39 show the ipcalc results for both addresses.

```
checkout@M2-secgrp3-11-lab2-HenriquesElgamal:~$ ipcalc 127.0.0.1
Address:   127.0.0.1           01111111.00000000.00000000. 00000001
Netmask:   255.255.255.0 = 24  11111111.11111111.11111111. 00000000
Wildcard:  0.0.0.255           00000000.00000000.00000000. 11111111
=>
Network:   127.0.0.0/24        01111111.00000000.00000000. 00000000
HostMin:   127.0.0.1           01111111.00000000.00000000. 00000001
HostMax:   127.0.0.254         01111111.00000000.00000000. 11111110
Broadcast: 127.0.0.255         01111111.00000000.00000000. 11111111
Hosts/Net: 254                  Class A, Loopback
```

Figure 38: ipcalc results showing that 127.0.0.1 belongs to the 127.0.0.0/8 local network.

```
checkout@M2-secgrp3-11-lab2-HenriquesElgamal:~$ ipcalc 127.0.1.1
Address:   127.0.1.1           01111111.00000000.00000001. 00000001
Netmask:   255.255.255.0 = 24  11111111.11111111.11111111. 00000000
Wildcard:  0.0.0.255           00000000.00000000.00000000. 11111111
=>
Network:   127.0.1.0/24        01111111.00000000.00000001. 00000000
HostMin:   127.0.1.1           01111111.00000000.00000001. 00000001
HostMax:   127.0.1.254         01111111.00000000.00000001. 11111110
Broadcast: 127.0.1.255         01111111.00000000.00000001. 11111111
Hosts/Net: 254                  Class A, Loopback

checkout@M2-secgrp3-11-lab2-HenriquesElgamal:~$ _
```

Figure 39: ipcalc results showing that 127.0.1.1 also belongs to the 127.0.0.0/8 local network.

SubA: (b) Zero-Configuration local Name Resolution (part of the coming task)

T2: Done. See the evidence (screenshots in images/L2E1S2/L2E1S2T2 subfolder and attachments, if any, in atts/L2E1S2/L2E1S2T2 subfolder.)

Q2: The question is

(a) Identify the two mdns packets (use frame numbers) that corresponds to the query (sent from M2) and the query response (sent from H2). Note that both are sent to 224.0.0.251.

(b) Who owns the IPv4 address 224.0.0.251?

(c) What transport layer protocol is used in the mdns frames?

(d) What are the dst and src transport port numbers used in the mdns query and query response?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A2: The answer is

(a) In our capture, Frame 1 was the IPv4 mDNS query sent from M2, and Frame 3 was the corresponding IPv4 mDNS response sent from H2. Both packets were sent to the multicast IPv4 address 224.0.0.251. Figure 40 shows the query, while Figure 41 shows the response.

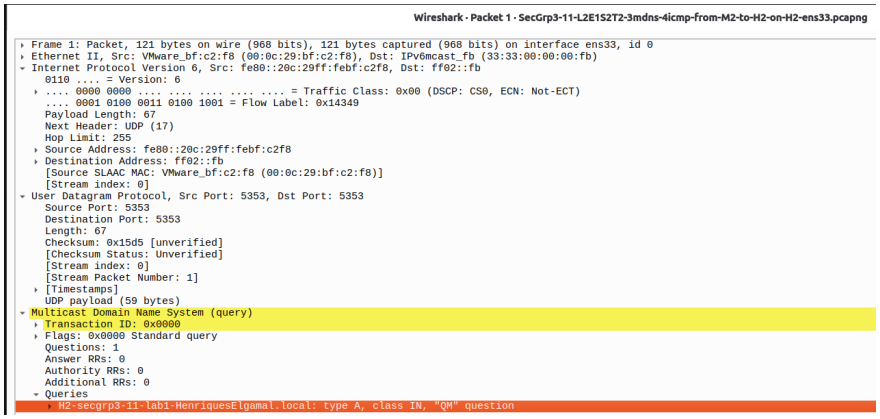


Figure 40: Ipv4 mDNS query from M2 to the 224.0.0.251 multicast address using UDP port 5353.

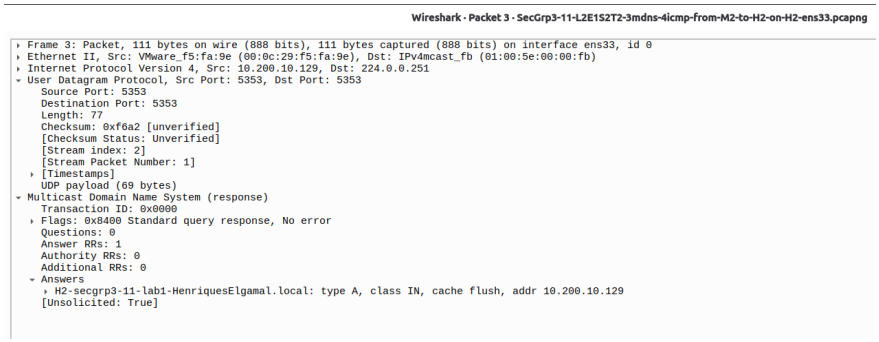


Figure 41: IPv4 mDNS response from H2 to the 224.0.0.251 multicast address using UDP port 5353.

(b) According to Gemini [5], 224.0.0.251 is not owned by one individual host. It is the multicast IPv4 address reserved for mDNS. Devices using mDNS send their queries and responses to this multicast group so that other mDNS-capable devices on the local network can receive them.

(c) The transport-layer protocol used by mDNS was UDP. Figures 42 and 43 show in the packet details.



Figure 42: mDNS query showing usage of Transport-layer protocol UDP.

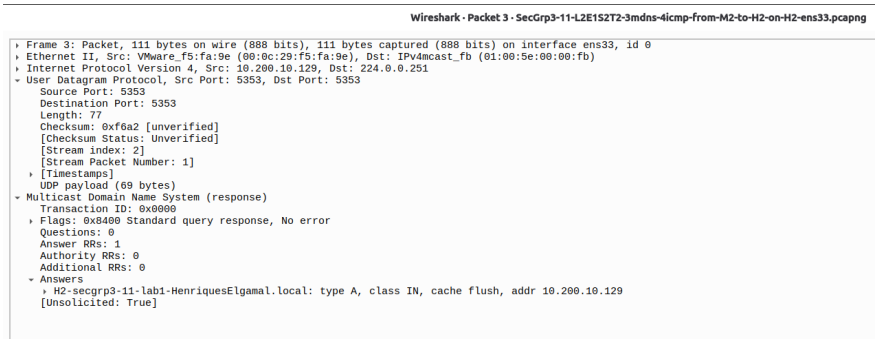


Figure 43: mDNS response showing usage of Transport-layer protocol UDP.

(d) The mDNS query used UDP source port 5353 and destination port 5353. The mDNS response also used source port 5353 and destination port 5353. These values are shown in the UDP headers of Frames 1 and 3

SubA: Simulate DNS Iterative Process

T3: Done. See the evidence (screenshots in images/L2E1S2/L2E1S2T3 subfolder and attachments, if any, in atts/L2E1S2/L2E1S2T3 subfolder.)

Q3: The question is

(a) How many dns queries are sent? How many answers (that is query responses) are received?

(b) What do the records in the "QUERY or QUESTION: 1", "ANSWER: 0","AUTHORITY: 13" and "ADDITIONAL: 26" sections represent?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A3: The answer is

(a) One DNS query was sent for www.jmu.ed and one DNS response was received. The response contained zero Answer records, meaning that the root DNS server did not directly return the IPv4 address of www.jmu.edu. Instead, it returned a referral to the .edu DNS servers.

(b) According to ChatGPT [6], and shown in Figure 44 QUERY: 1 means that the DNS message contained one question for www.jmu.edu. ANSWER: 0 means there was no final answer record for www.jmu.edu in this response. AUTHORITY: 13 means the response contained 13 name-server records identifying the DNS servers responsible for the .edu top-level domain. ADDITIONAL: 26 contains additional address records that provide IPv4 and/or IPv6 addresses for those name servers, allowing the next DNS server to be contacted.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @198.41.0.4 +nodnssec +noedn
s -4 www.jmu.edu
;; Truncated, retrying in TCP mode.

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @198.41.0.4 +nodnssec +noedns -4
 www.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44608
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 13, ADDITIONAL: 26
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;www.jmu.edu.                   IN      A

;; AUTHORITY SECTION:
edu.                    172800  IN      NS      d.edu-servers.net.
edu.                    172800  IN      NS      b.edu-servers.net.
edu.                    172800  IN      NS      f.edu-servers.net.
edu.                    172800  IN      NS      h.edu-servers.net.
```

Figure 44: Root DNS query for www.jmu.edu showing one question, no final answer, and a referall to the .edu name servers

Q4: The question is

(a) How many dns queries are sent? How many answers (that is query responses) are received?

(b) What do the records in the "QUERY or QUESTION: 1", "ANSWER: 0", "AUTHORITY: 3" and "ADDITIONAL: 3" sections represent?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A4: The answer is

(a) Shown by Figure 45, One DNS query was sent to the .edu DNS server at 192.5.6.30 and one DNS response was received. The response contained zero Answer records, because the .edu server did not provide the final IPv4 address for www.jmu.edu. Instead, it referred the query to the authoritative DNS servers for jmu.edu.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @192.5.6.30 +nodnssec +noedn
s -4 www.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @192.5.6.30 +nodnssec +noedns -4
 www.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6742
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;www.jmu.edu.                   IN      A

;; AUTHORITY SECTION:
jmu.edu.                172800  IN      NS      ns3.jmu.edu.
jmu.edu.                172800  IN      NS      ns4.jmu.edu.
jmu.edu.                172800  IN      NS      ns5.jmu.edu.

;; ADDITIONAL SECTION:
ns3.jmu.edu.            172800  IN      A       134.126.115.46
ns4.jmu.edu.            172800  IN      A       134.126.115.45
ns5.jmu.edu.            172800  IN      A       34.199.120.69

;; Query time: 59 msec
;; SERVER: 192.5.6.30#53(192.5.6.30) (UDP)
;; WHEN: Sat Sep 26 18:40:04 EDT 2026
;; MSG SIZE  rcvd: 131

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 45: DNS query to the .edu TLD server at 192.5.6.30 showing no final answer and a referral to the three authoritative jmu.edu name servers

(b) Shown by Figure 45, QUERY: 1 represents the single question requesting information about www.jmu.edu. ANSWER: 0 means the .edu server did not provide the final host address. AUTHORITY: 3 represents the three authoritative name-server records for the jmu.edu domain. ADDITIONAL: 3 contains the IPv4 addresses needed to contact those three JMU name servers

Q5: The question is

(a) How many dns queries are sent? How many answers (that is query responses) are received?

(b) What do the records in the "QUERY or QUESTION: 1", "ANSWER: 1", "AUTHORITY: 3" and "ADDITIONAL: 3" sections represent?

(c) What is the IPv4 address of www.jmu.edu?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A5: The answer is

(a) According to Figure 46, One DNS question was sent to the JMU authoritative DNS server at 134.126.115.46 and one DNS response was received. This response contained one Answer record, meaning that the authoritative server returned the requested IPv4 address for www.jmu.edu.

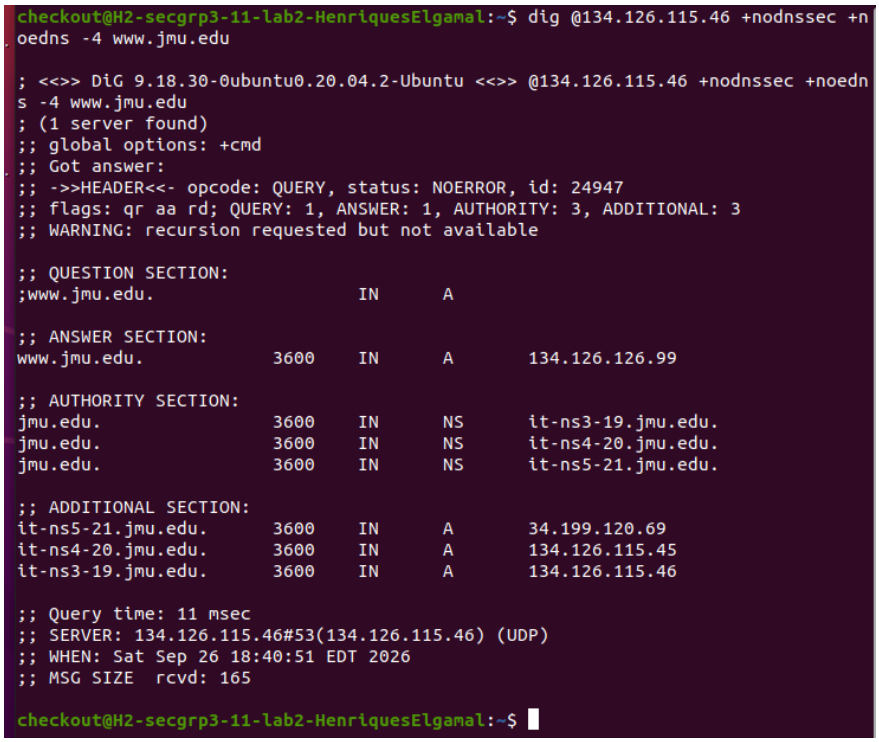


Figure 46: Query to the JMU authoritative DNS server showing 134.126.126.99 as the resolved IPv4 address for www.jmu.edu

(b) QUERY: 1 represents the single DNS question for www.jmu.edu. ANSWER: 1 represents the A record containing the resolved IPv4 address of www.jmu.edu. AUTHORITY: 3 identifies the three authoritative name servers for jmu.edu, while ADDITIONAL: 3 provides additional address information for those name servers.

(c) The IPv4 address returned for www.jmu.edu in our experiment was: 134.126.126.99 shown by Figure 47

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @134.126.115.46 +nodnssec +n
oedns -4 www.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @134.126.115.46 +nodnssec +noedn
s -4 www.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24947
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;www.jmu.edu.                   IN      A

;; ANSWER SECTION:
www.jmu.edu.            3600    IN      A       134.126.126.99

;; AUTHORITY SECTION:
jmu.edu.                3600    IN      NS      it-ns3-19.jmu.edu.
jmu.edu.                3600    IN      NS      it-ns4-20.jmu.edu.
jmu.edu.                3600    IN      NS      it-ns5-21.jmu.edu.

;; ADDITIONAL SECTION:
it-ns5-21.jmu.edu.      3600    IN      A       34.199.120.69
it-ns4-20.jmu.edu.      3600    IN      A       134.126.115.45
it-ns3-19.jmu.edu.      3600    IN      A       134.126.115.46

;; Query time: 11 msec
;; SERVER: 134.126.115.46#53(134.126.115.46) (UDP)
;; WHEN: Sat Sep 26 18:40:51 EDT 2026
;; MSG SIZE  rcvd: 165

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 47: Query to the JMU authoritative DNS server showing 134.126.126.99 as the resolved IPv4 address for www.jmu.edu

Q6: The question is

(a) Provide the resolved IPv4 address(es).

(b) Show all your steps with clear evidence.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A6: The answer is

(a) The resolved IPv4 address for www.jmu.edu is 134.126.126.99, and the resolved IPv4 address for stu.cs.jmu.edu is 134.126.141.221. Figures 48 and 49 show the final DNS responses containing these resolved IPv4 addresses.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @134.126.115.46 +nodnssec +n
oedns -4 www.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @134.126.115.46 +nodnssec +noedn
s -4 www.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24947
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;www.jmu.edu.                   IN      A

;; ANSWER SECTION:
www.jmu.edu.            3600    IN      A       134.126.126.99

;; AUTHORITY SECTION:
jmu.edu.                3600    IN      NS      it-ns3-19.jmu.edu.
jmu.edu.                3600    IN      NS      it-ns4-20.jmu.edu.
jmu.edu.                3600    IN      NS      it-ns5-21.jmu.edu.

;; ADDITIONAL SECTION:
it-ns5-21.jmu.edu.      3600    IN      A       34.199.120.69
it-ns4-20.jmu.edu.      3600    IN      A       134.126.115.45
it-ns3-19.jmu.edu.      3600    IN      A       134.126.115.46

;; Query time: 11 msec
;; SERVER: 134.126.115.46#53(134.126.115.46) (UDP)
;; WHEN: Sat Sep 26 18:40:51 EDT 2026
;; MSG SIZE  rcvd: 165

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 48: Final authoritative DNS query showing www.jmu.edu resolved to IPv4 address 134.126.126.99.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @134.126.115.46 +nodnssec +n
oedns -4 stu.cs.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @134.126.115.46 +nodnssec +noedn
s -4 stu.cs.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23941
;; flags: qr aa rd; QUERY: 1, ANSWER: 2, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;stu.cs.jmu.edu.                        IN      A

;; ANSWER SECTION:
stu.cs.jmu.edu.         3600    IN      CNAME   student.cs.jmu.edu.
student.cs.jmu.edu.     3600    IN      A       134.126.141.221

;; AUTHORITY SECTION:
cs.jmu.edu.             3600    IN      NS      it-ns5-21.jmu.edu.
cs.jmu.edu.             3600    IN      NS      it-ns3-19.jmu.edu.
cs.jmu.edu.             3600    IN      NS      it-ns4-20.jmu.edu.

;; ADDITIONAL SECTION:
it-ns5-21.jmu.edu.      3600    IN      A       34.199.120.69
it-ns4-20.jmu.edu.      3600    IN      A       134.126.115.45
it-ns3-19.jmu.edu.      3600    IN      A       134.126.115.46

;; Query time: 12 msec
;; SERVER: 134.126.115.46#53(134.126.115.46) (UDP)
;; WHEN: Sat Sep 26 18:53:23 EDT 2026
;; MSG SIZE  rcvd: 190

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 49: Final authoritative DNS query showing stu.cs.jmu.edu as a CNAME for student.cs.jmu.edu with resolved IPv4 address 134.126.141.221.

(b) We resolved the addresses by following the DNS hierarchy with dig. We first queried a root DNS server, which returned the .edu DNS servers, as shown in Figure 50. We then queried the .edu DNS server at 192.5.6.30, which returned the JMU authoritative DNS servers, as shown in Figures 51 and 52. Finally, we queried the JMU authoritative DNS server at 134.126.115.46. Figure 48 shows that www.jmu.edu resolves to 134.126.126.99, and Figure 53 shows that stu.cs.jmu.edu is a CNAME for student.cs.jmu.edu, which resolves to 134.126.141.221.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @198.41.0.4 +nodnssec +noedn
s -4 www.jmu.edu
;; Truncated, retrying in TCP mode.

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @198.41.0.4 +nodnssec +noedns -4
 www.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44608
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 13, ADDITIONAL: 26
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;www.jmu.edu.                   IN      A

;; AUTHORITY SECTION:
edu.                    172800  IN      NS      d.edu-servers.net.
edu.                    172800  IN      NS      b.edu-servers.net.
edu.                    172800  IN      NS      f.edu-servers.net.
edu.                    172800  IN      NS      h.edu-servers.net.
```

Figure 50: Root DNS query for www.jmu.edu showing the referral to the .edu DNS servers.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @192.5.6.30 +nodnssec +noedn
s -4 www.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @192.5.6.30 +nodnssec +noedns -4
 www.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6742
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;www.jmu.edu.                   IN      A

;; AUTHORITY SECTION:
jmu.edu.                172800  IN      NS      ns3.jmu.edu.
jmu.edu.                172800  IN      NS      ns4.jmu.edu.
jmu.edu.                172800  IN      NS      ns5.jmu.edu.

;; ADDITIONAL SECTION:
ns3.jmu.edu.            172800  IN      A       134.126.115.46
ns4.jmu.edu.            172800  IN      A       134.126.115.45
ns5.jmu.edu.            172800  IN      A       34.199.120.69

;; Query time: 59 msec
;; SERVER: 192.5.6.30#53(192.5.6.30) (UDP)
;; WHEN: Sat Sep 26 18:40:04 EDT 2026
;; MSG SIZE  rcvd: 131

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 51: Query to the .edu DNS server at 192.5.6.30 showing the JMU authoritative DNS servers for www.jmu.edu.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @192.5.6.30 +nodnssec +noedn
s -4 stu.cs.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @192.5.6.30 +nodnssec +noedns -4
 stu.cs.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 54068
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;stu.cs.jmu.edu.                        IN      A

;; AUTHORITY SECTION:
jmu.edu.                172800  IN      NS      ns3.jmu.edu.
jmu.edu.                172800  IN      NS      ns4.jmu.edu.
jmu.edu.                172800  IN      NS      ns5.jmu.edu.

;; ADDITIONAL SECTION:
ns3.jmu.edu.            172800  IN      A       134.126.115.46
ns4.jmu.edu.            172800  IN      A       134.126.115.45
ns5.jmu.edu.            172800  IN      A       34.199.120.69

;; Query time: 52 msec
;; SERVER: 192.5.6.30#53(192.5.6.30) (UDP)
;; WHEN: Sat Sep 26 18:51:27 EDT 2026
;; MSG SIZE  rcvd: 134

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 52: Query to the .edu DNS server at 192.5.6.30 showing the JMU authoritative DNS servers for stu.cs.jmu.edu.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ dig @134.126.115.46 +nodnssec +n
oedns -4 stu.cs.jmu.edu

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> @134.126.115.46 +nodnssec +noedn
s -4 stu.cs.jmu.edu
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23941
;; flags: qr aa rd; QUERY: 1, ANSWER: 2, AUTHORITY: 3, ADDITIONAL: 3
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;stu.cs.jmu.edu.                        IN      A

;; ANSWER SECTION:
stu.cs.jmu.edu.         3600    IN      CNAME   student.cs.jmu.edu.
student.cs.jmu.edu.     3600    IN      A       134.126.141.221

;; AUTHORITY SECTION:
cs.jmu.edu.             3600    IN      NS      it-ns5-21.jmu.edu.
cs.jmu.edu.             3600    IN      NS      it-ns3-19.jmu.edu.
cs.jmu.edu.             3600    IN      NS      it-ns4-20.jmu.edu.

;; ADDITIONAL SECTION:
it-ns5-21.jmu.edu.      3600    IN      A       34.199.120.69
it-ns4-20.jmu.edu.      3600    IN      A       134.126.115.45
it-ns3-19.jmu.edu.      3600    IN      A       134.126.115.46

;; Query time: 12 msec
;; SERVER: 134.126.115.46#53(134.126.115.46) (UDP)
;; WHEN: Sat Sep 26 18:53:23 EDT 2026
;; MSG SIZE  rcvd: 190

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 53: Final authoritative DNS query showing stu.cs.jmu.edu as a CNAME for student.cs.jmu.edu with resolved IPv4 address 134.126.141.221.

SubA: DNS Client (dynamically assigned a gateway)

T4: Done. See the evidence (screenshots in images/L2E1S2/L2E1S2T4 subfolder and attachments, if any, in atts/L2E1S2/L2E1S2T4 subfolder.)

Q7: The question is

(a) Compare the results of the two methods and provide the results.

(b) Does the DNS IPv4 address assigned represent the IPv4 address of a DNS server that resolves domain name?

(c) if not, who owns that IPv4 address?

(d) Is that IPv4 address assigned statically or dynamically and by who?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A7: The answer is

(a) Both methods produced the same result. The graphical Wired settings and the resolvectl output both show that the DNS IPv4 address used by H2 is 10.200.10.2. Figure 54 shows the Current DNS Server and

DNS Servers as 10.200.10.2, and Figure 55 shows DNS as 10.200.10.2 in the Wired network settings.

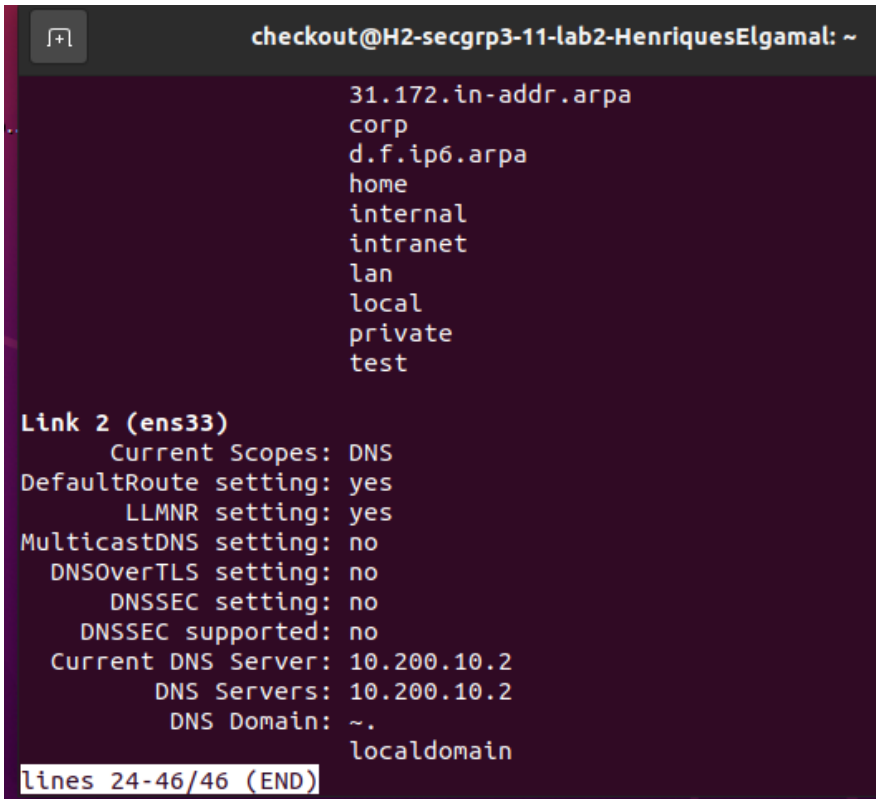


Figure 54: resolvectl output showing that H2 is using 10.200.10.2 as its current DNS server.

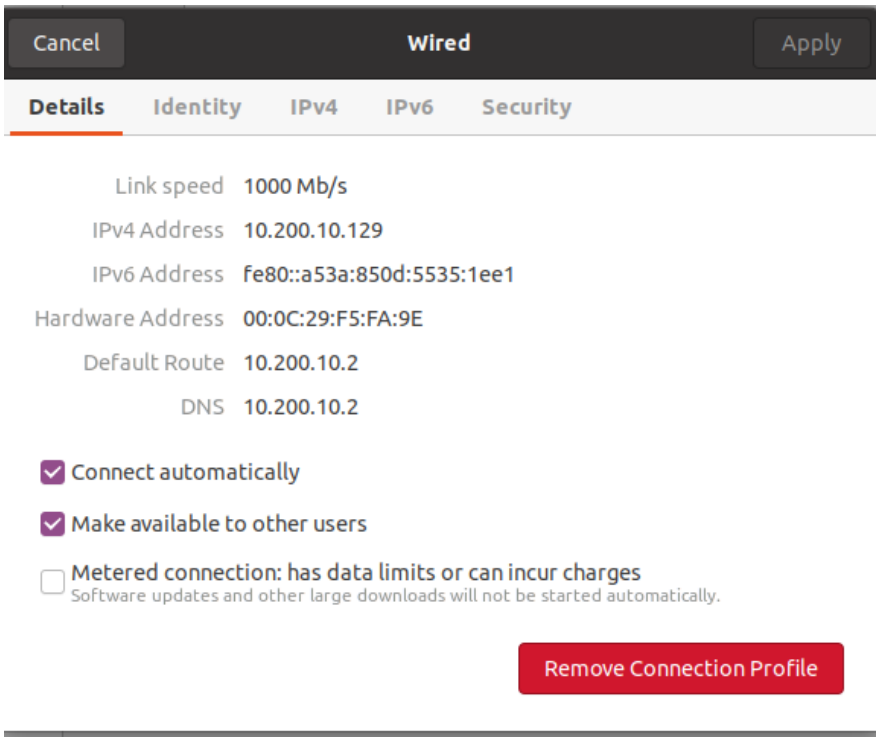


Figure 55: H2 Wired network settings showing IPv4 address 10.200.10.129 and 10.200.10.2 as the default route and DNS address.

(b) No. The DNS IPv4 address 10.200.10.2 is being used by H2 as its DNS resolver. Figure 56 confirms that H2 at 10.200.10.129 sends a DNS query to 10.200.10.2 using destination port 53.

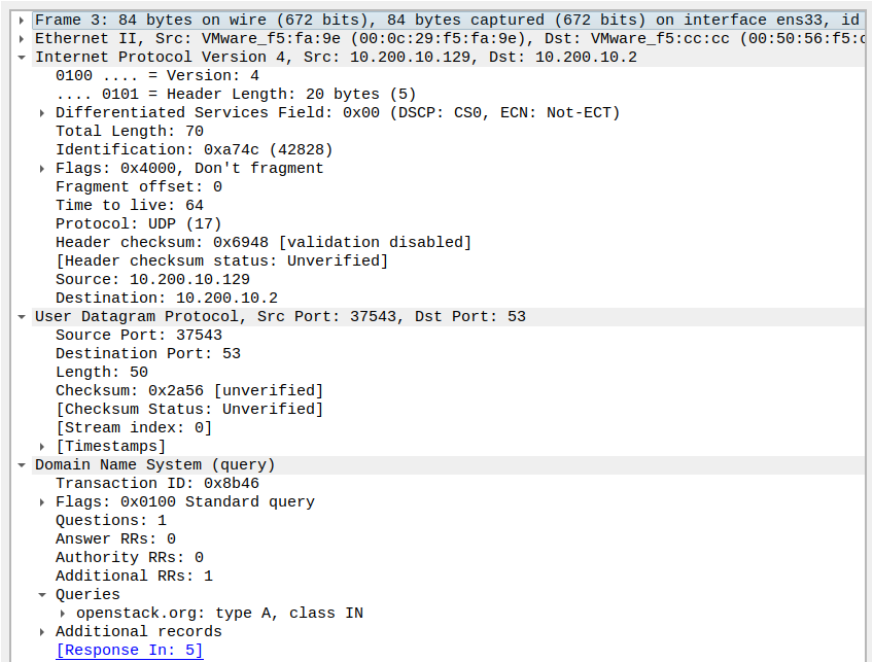


Figure 56: Wireshark DNS query details showing H2 at 10.200.10.129 sending UDP traffic to DNS resolver 10.200.10.2 on destination port 53.

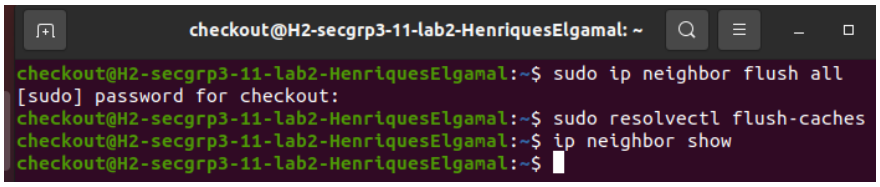


Figure 57: Wireshark DNS response showing openstack.org resolved to IPv4 addresses 172.66.40.143 and 172.66.43.113.

(c) The IPv4 address 10.200.10.2 belongs to the VMware virtual network infrastructure and acts as the DNS resolver for H2

(d) The DNS IPv4 address was assigned dynamically to H2 through the VMware virtual network configuration. Figure 58 shows H2 using 10.200.10.2 as both its Default Route and DNS address.

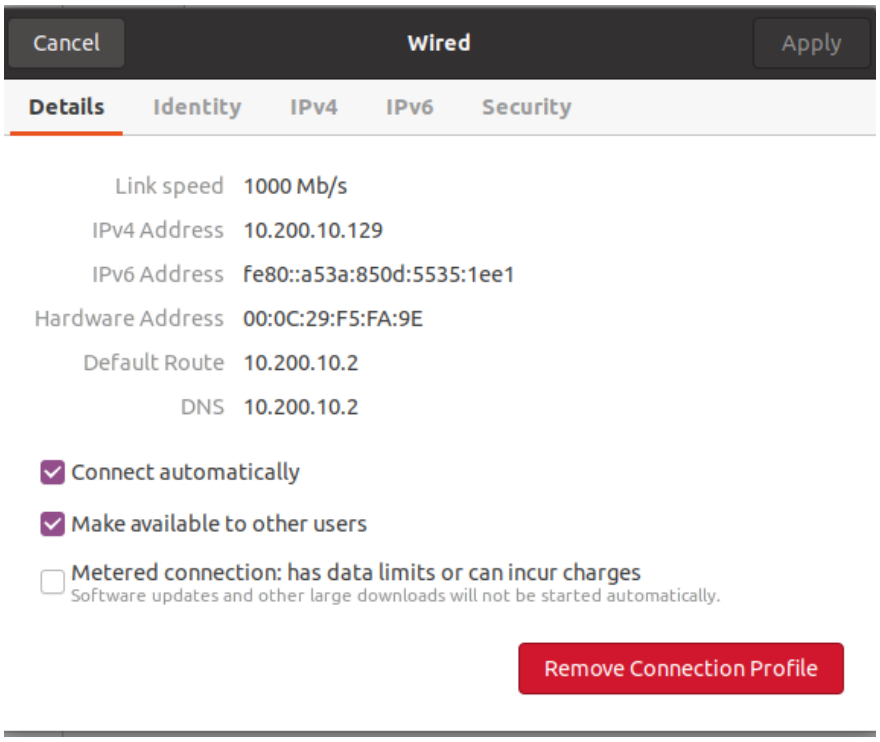


Figure 58: Network settings showing H2 at 10.200.10.129 with Default Route and DNS set to 10.200.10.2.

Q8: The question is

(a) Explain why the above ping command triggered the exchange of these 3 sets of packets.

(b) What are the differences between query for A and AAAA records?

(c) Who initiated the dns query (src), to whom (dst) and to what transport port number (dst)?

(d) What is(are) the IPv4 address(s) provided in the query response (that is the resolved IPv4 addresses) for IPv4?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A8: The answer is

(a) The ping command triggered three sets of packets because H2 first needed to find the EUI-48 address associated with 10.200.10.2 using ARP. H2 then sent DNS queries to resolve the domain name openstack.org. After the IPv4 address was resolved, H2 sent the ICMP Echo Request and received an ICMP Echo Reply. Figure 59 shows these three sets of packets: ARP, DNS, and ICMP.

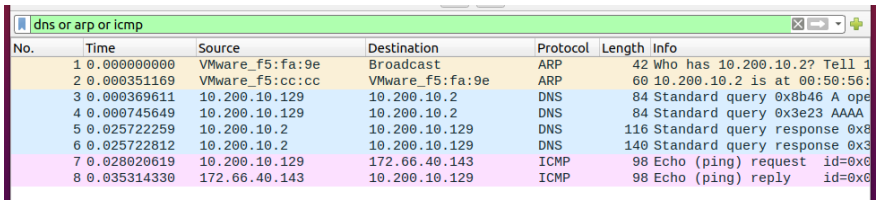


Figure 59: Wireshark capture showing the ARP, DNS, and ICMP packets triggered by the ping command.

(b) According to EasyDMarc [7], An A record query asks DNS for the IPv4 address of a domain name, while an AAAA record query asks for its IPv6 address. Figure 60 shows an A query for openstack.org, and Figure 61 shows an AAAA query for openstack.org.

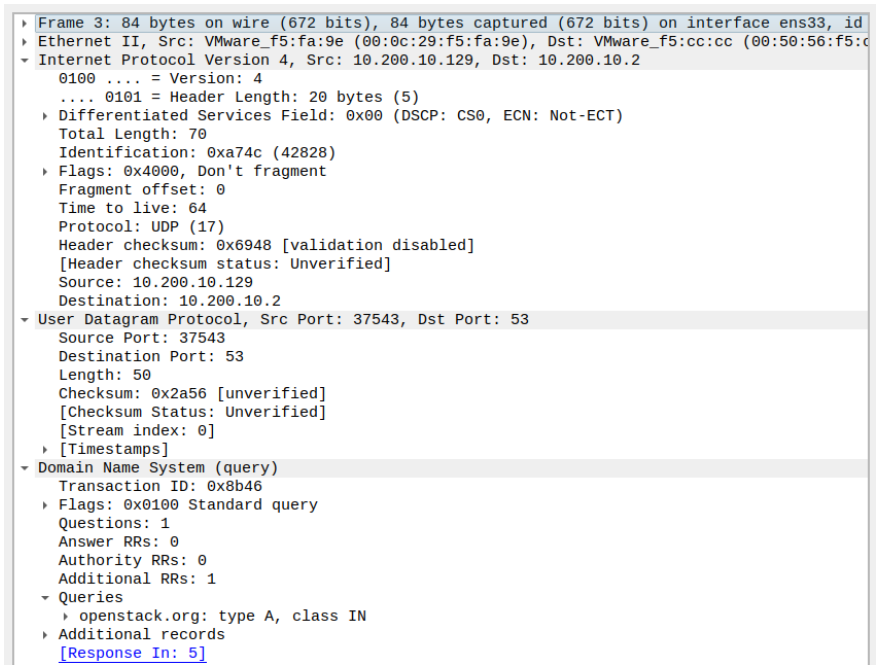


Figure 60: Wireshark details of the DNS A record query from H2 at 10.200.10.129 to 10.200.10.2 using UDP destination port 53.

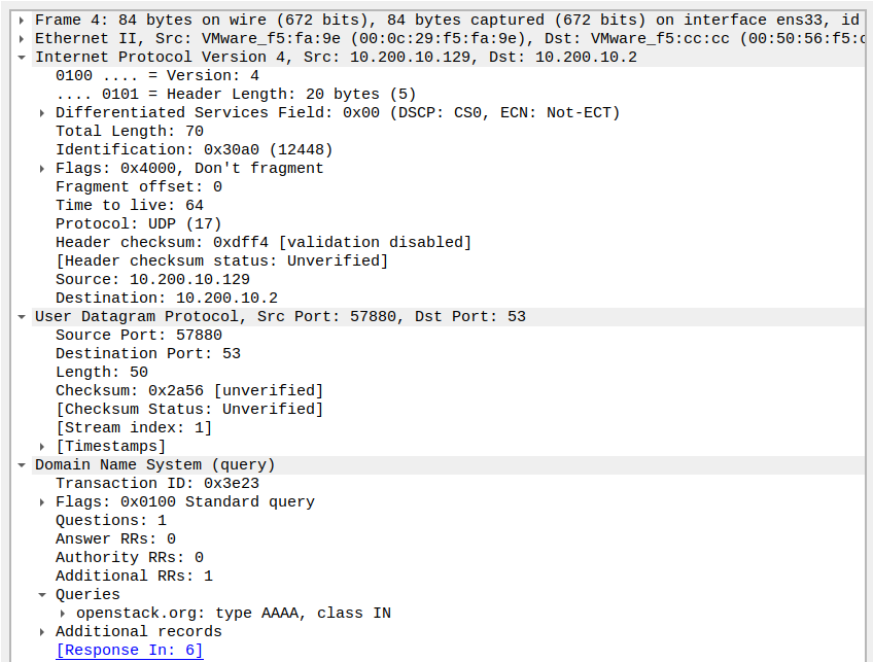


Figure 61: Wireshark details showing the AAAA record query for the IPv6 address of openstack.org.

(c) H2 initiated the DNS query. The source IPv4 address was 10.200.10.129 and the destination IPv4 address was 10.200.10.2. The destination transport port was UDP port 53. Figure 62 shows the source, destination, and UDP destination port 53.

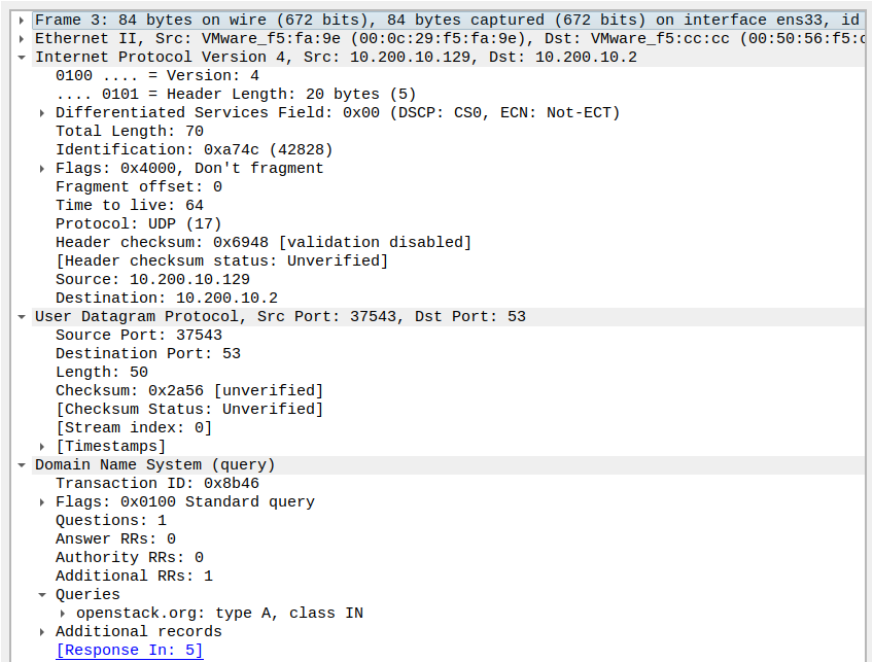


Figure 62: Wireshark details of the DNS A record query from H2 at 10.200.10.129 to 10.200.10.2 using UDP destination port 53.

(d) The IPv4 addresses provided in the DNS query response for openstack.org were 172.66.40.143 and 172.66.43.113. Figure 63 shows both IPv4 addresses in the DNS Answer section.

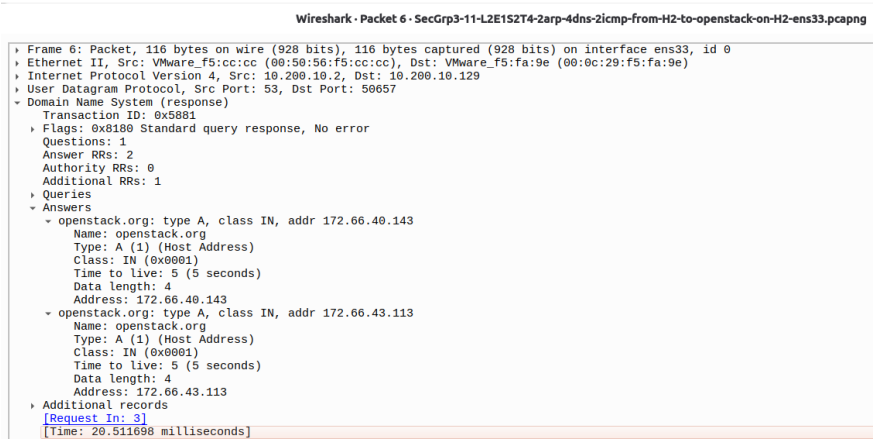


Figure 63: Wireshark DNS response showing the resolved IPv4 addresses 172.66.40.143 and 172.66.43.113 for openstack.org.

Q9: The question is

(a) Explain why the dns command triggered the exchange of these 2 sets of packets.

(b) Compare the DNS resolver IPv4 used in the above two cases based on the command line response and the packets captured. Hint: 127.0.0.53, 10.200.10.2. Hint: compare means multiple evidence.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A9: The answer is

(a) The nslookup command triggered two sets of packets because H2 first used ARP to find the EUI-48 address associated with 10.200.10.2. H2 then exchanged DNS packets with 10.200.10.2 to resolve openstack.org. Figure 64 shows the ARP and DNS packets generated by the nslookup command.

dns or arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	VMware_f5:fa:9e	Broadcast	ARP	42	Who has 10.200.10.2? Tel
2	0.000409915	VMware_f5:cc:cc	VMware_f5:fa:9e	ARP	60	10.200.10.2 is at 00:50:
3	0.000471254	10.200.10.129	10.200.10.2	DNS	84	Standard query 0x73b9 A
4	0.024401029	10.200.10.2	10.200.10.129	DNS	116	Standard query response
5	0.025601469	10.200.10.129	10.200.10.2	DNS	84	Standard query 0x2677 AA
6	0.045974015	10.200.10.2	10.200.10.129	DNS	140	Standard query response

Figure 64: Wireshark capture showing the ARP and DNS packets generated during the nslookup command, with DNS traffic between H2 at 10.200.10.129 and 10.200.10.2.

(b) The command-line response and the packet capture show two DNS resolver IPv4 addresses because they represent different parts of the DNS resolution process. The nslookup output in Figure 66 shows 127.0.0.53 as the DNS server used locally by H2. However, the Wireshark capture in Figure 65 shows that the DNS packets sent onto the network go from H2 at 10.200.10.129 to 10.200.10.2. Figure 67 further confirms that the DNS query has source IPv4 address 10.200.10.129, destination IPv4 address 10.200.10.2, and UDP destination port 53. Therefore, 127.0.0.53 is the local DNS stub resolver on H2, while 10.200.10.2 is the DNS resolver contacted over the network.

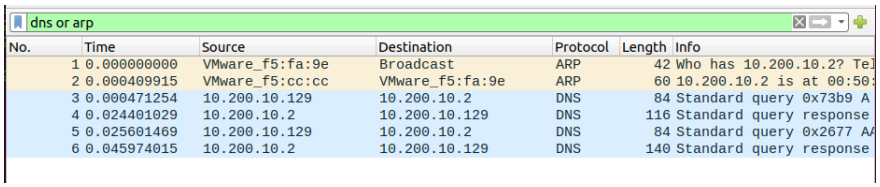


Figure 65: Wireshark capture showing the ARP and DNS packets generated during the nslookup command, with DNS traffic between H2 at 10.200.10.129 and 10.200.10.2.

```
checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$ nslookup openstack.org
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   openstack.org
Address: 172.66.40.143
Name:   openstack.org
Address: 172.66.43.113
Name:   openstack.org
Address: 2606:4700:3108::ac42:2b71
Name:   openstack.org
Address: 2606:4700:3108::ac42:288f

checkout@H2-secgrp3-11-lab2-HenriquesElgamal:~$
```

Figure 66: nslookup command-line output showing 127.0.0.53 as the local DNS resolver and the resolved addresses for openstack.org.

```
▾ Internet Protocol Version 4, Src: 10.200.10.129, Dst: 10.200.10.2
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 70
    Identification: 0x1d48 (7496)
  ▸ Flags: 0x4000, Don't fragment
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0xf34c [validation disabled]
    [Header checksum status: Unverified]
    Source: 10.200.10.129
    Destination: 10.200.10.2
▾ User Datagram Protocol, Src Port: 60232, Dst Port: 53
    Source Port: 60232
    Destination Port: 53
    Length: 50
    Checksum: 0x2a56 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
  ▸ [Timestamps]
```

Figure 67: Wireshark DNS query details showing H2 at 10.200.10.129 sending the DNS query to 10.200.10.2 using UDP destination port 53.

SubA: DNS Client (statically assigned a specific DNS resolver)

T5: Done. See the evidence (screenshots in images/L2E1S2/L2E1S2T5 subfolder and attachments, if any, in atts/L2E1S2/L2E1S2T5 subfolder.)

Q10: The question is

(a) Who initiated the dns query (src) and to whom (dst)?

(b) Is the dns query recipient a public or private dns resolver?

(c) What is the transport layer protocol of the dns query? What are transport port numbers of the dns query?

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A10: The answer is

(a) H2 initiated the DNS query from 10.200.10.129 and sent it to 8.8.8.8. Figure 68 shows the DNS traffic between H2 and 8.8.8.8.

dns or arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	VMware_f5:fa:9e	Broadcast	ARP	42	Who has 10.200.10.2? Tell 10.200.10.129
2	0.000300605	VMware_f5:cc:cc	VMware_f5:fa:9e	ARP	60	10.200.10.2 is at 00:50:56:f5:cc:cc
3	0.000315677	10.200.10.129	8.8.8.8	DNS	84	Standard query 0x39ff A openstack.org OPT
4	0.021012599	8.8.8.8	10.200.10.129	DNS	116	Standard query response 0x39ff A openstack.org A 172.6
5	0.023103189	10.200.10.129	8.8.8.8	DNS	84	Standard query 0xd8b8 AAAA openstack.org OPT
6	0.041063267	8.8.8.8	10.200.10.129	DNS	140	Standard query response 0xd8b8 AAAA openstack.org AAAA

Figure 68: Wireshark capture showing DNS queries between H2 at 10.200.10.129 and the DNS resolver at 8.8.8.8.

(b) The recipient, 8.8.8.8, is a public DNS resolver.

(c) The DNS query uses UDP. Figure 69 shows the source port is 32868 and the destination port is 53.

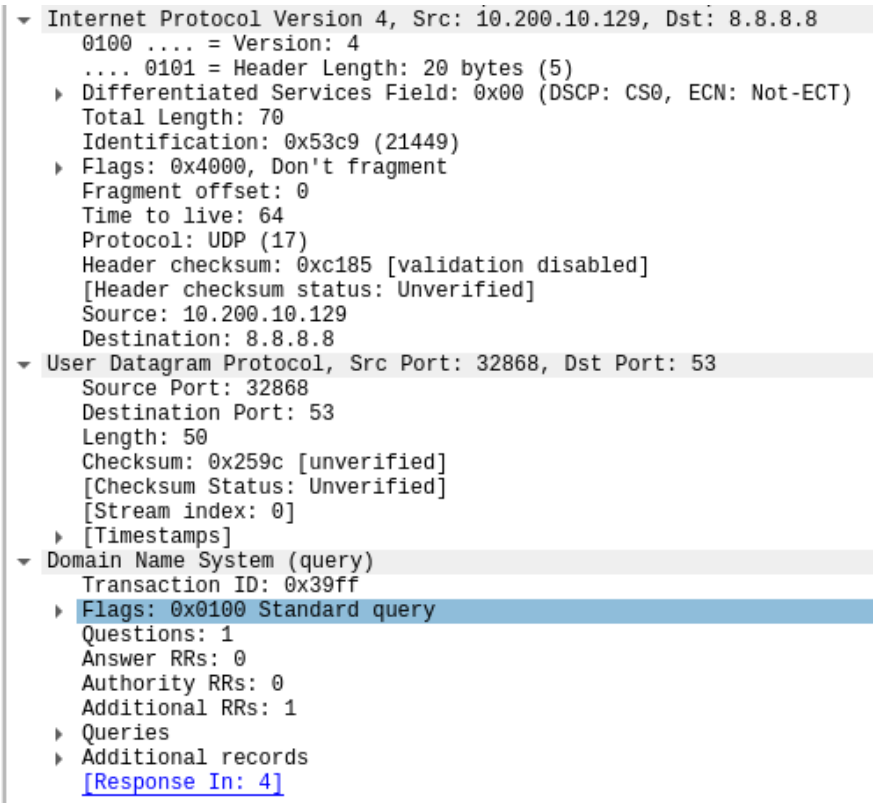


Figure 69: DNS query details showing UDP from 10.200.10.129 to 8.8.8.8 with source port 32868 and destination port 53.

3.1.3 Step 3: DHCP Server and DHCP Relay for multiple subnets environment

Figure 3 represents the network we used to perform all or some of the tasks in Step 3.

SubA: Introduction:

T1: Done. See the evidence (screenshots in images/L2E1S3/L2E1S3T1 subfolder and attachments, if any, in atts/L2E1S3/L2E1S3T1 subfolder.)

T2: Done. See the evidence (screenshots in images/L2E1S3/L2E1S3T2 subfolder and attachments, if any, in atts/L2E1S3/L2E1S3T2 subfolder.)

Q1: The question is
Why are we using ss with nltu options instead of the typical option nlt? Provide reference to evidence including screenshots, AI Apps (e.g., AI Apps

(e.g., ChatGPT)) conversations (attachment) or sharing URL. Reference the evidence with specific description of the relevant information. Provide citations to external sources.
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A1: The answer is
We used ss -a -nltu because the u option shows UDP ports in addition to TCP ports. DHCP uses UDP [8],, so this allows us to see the DHCP port. Figure 70 shows the UDP ports using the ss -a -nltu command.

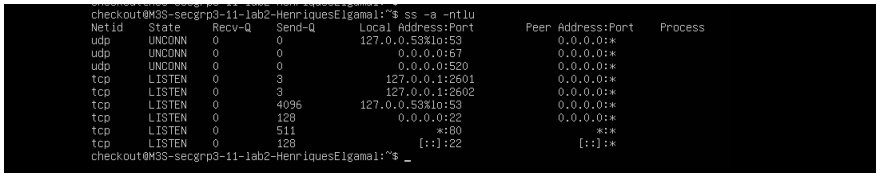


Figure 70: DHCP server showing UDP ports using the ss -a -nltu command.

T3: Done. See the evidence (screenshots in images/L2E1S3/L2E1S3T3 subfolder and attachments, if any, in atts/L2E1S3/L2E1S3T3 subfolder.)

Q2: The question is

(a) Was the connectivity test successful?

(b) Compare the results (IPv4 address perspective) of (4 sources) (1) the ip a on H3S, (2) the IPv4 address offered by the DHCP server and requested by the DHCP client as provided by the dhcp packets, (3) the dhcpd.leases file and (4) the results of dhcp-lease-list on M3S.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A2: The answer is

(a) Yes, the connectivity test was successful. H3S was able to ping M3S and received all 4 replies with 0% packet loss. Figure 71 shows the successful ping.

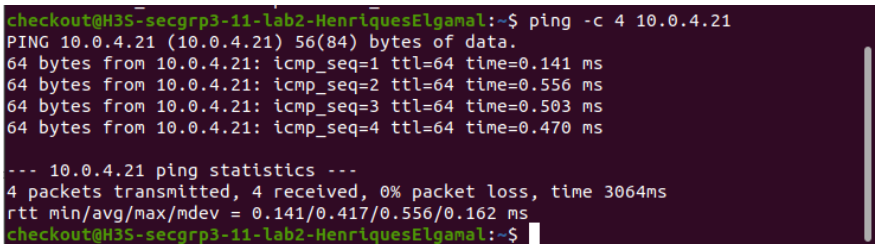


Figure 71: Successful connectivity test between H3S and M3S.

(b) The results were consistent between the different sources. The DHCP packets show that H3S was offered and requested the IPv4 address 10.0.4.11. The dhcpd.leases file and dhcp-lease-list also show 10.0.4.11 for H3S. Figures 72, 73, and 74 show these results.

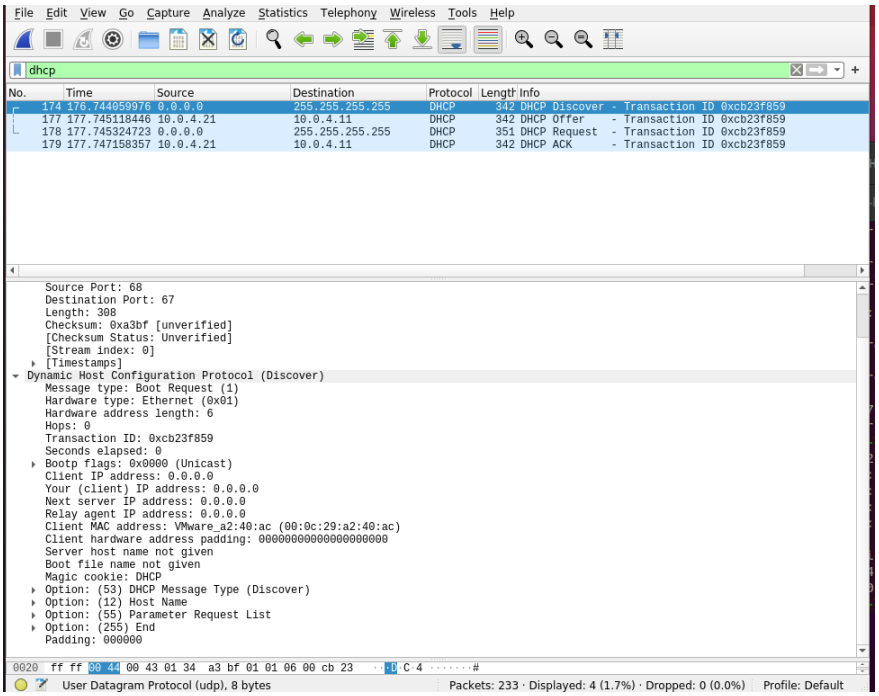


Figure 72: DHCP packets showing the IPv4 address offered to H3S.

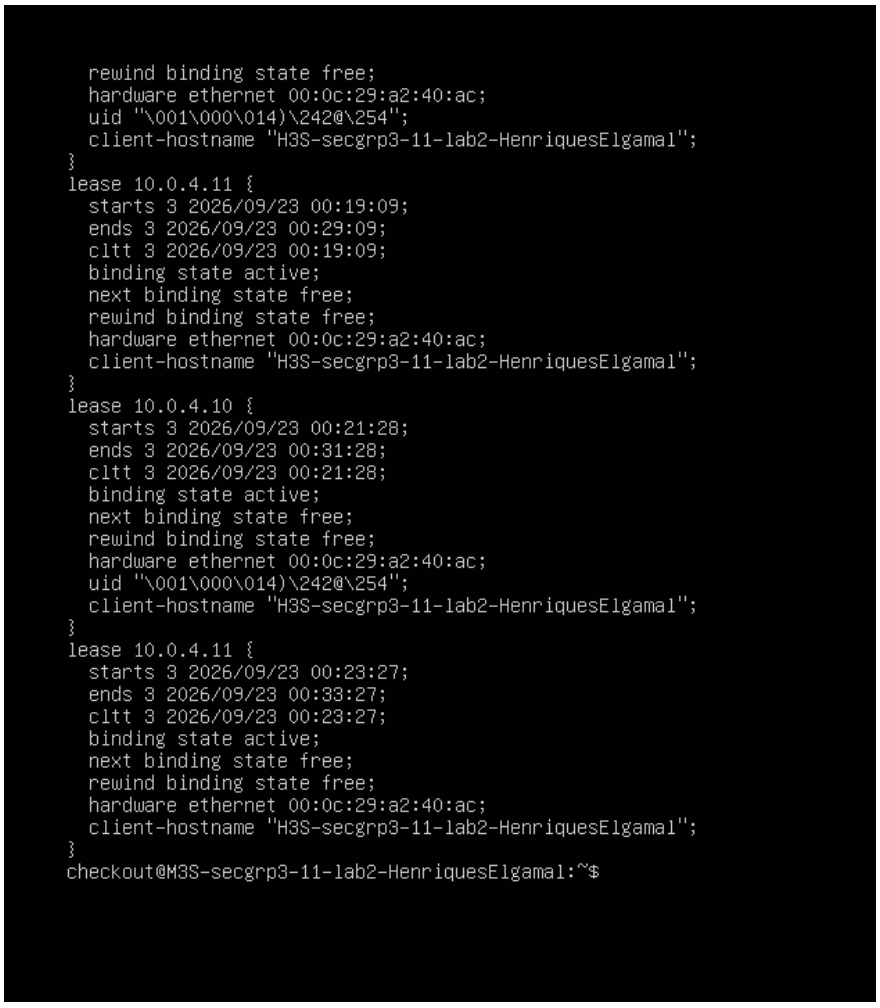


Figure 73: DHCP server lease file showing the IPv4 address assigned to H3S.

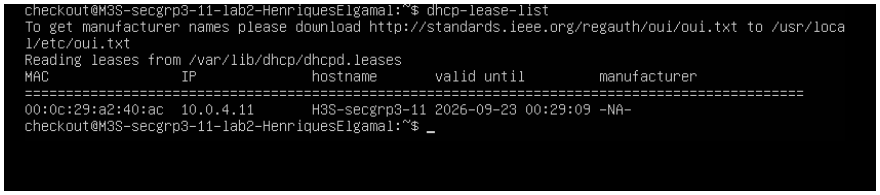


Figure 74: DHCP lease list showing the IPv4 address assigned to H3S.

Q3: The question is

(a) Find out the protocol used by DHCP at the transport layer.

(b) Find out the transport port numbers used by the DHCP client and server.

(c) Explain the payload of the DHCP offer packet.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A3: The answer is

(a) DHCP uses UDP at the transport layer. Figure 75 shows the DHCP packets using UDP.

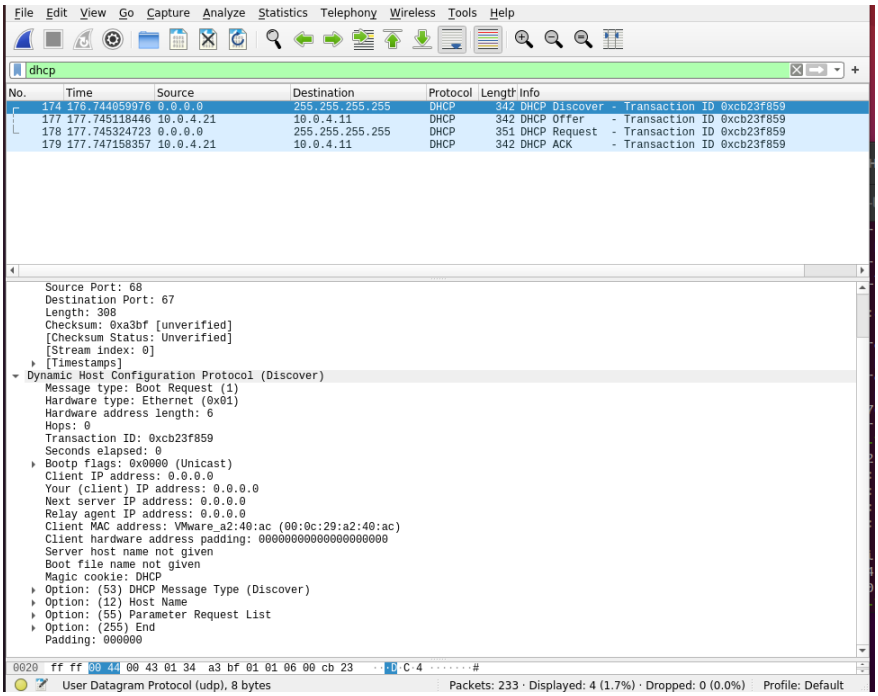


Figure 75: DHCP packets showing the UDP transport information and port numbers.

(b) The DHCP client uses UDP port 68 and the DHCP server uses UDP port 67 [9]. Figure 76 shows the transport port information.

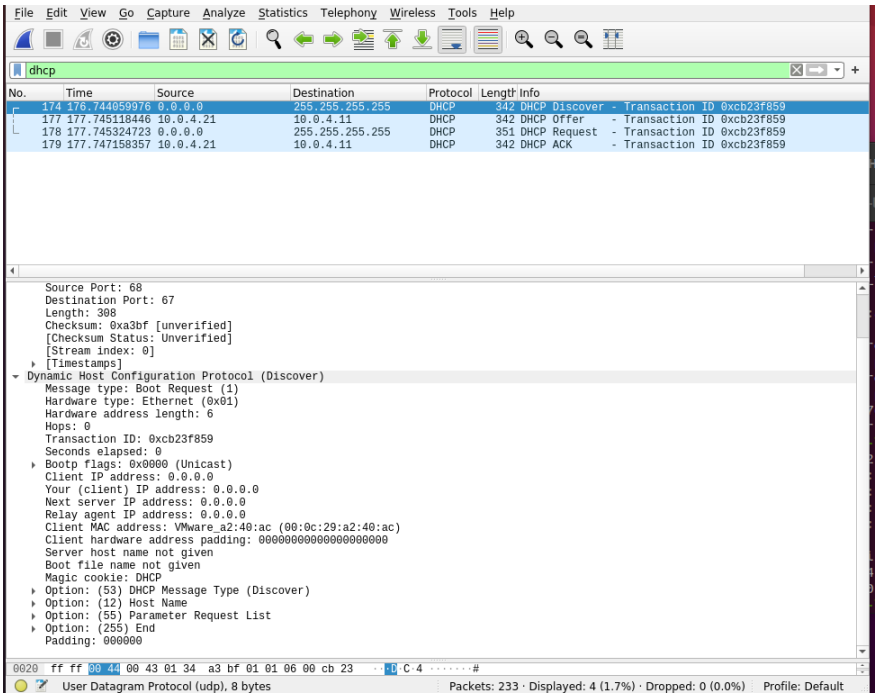


Figure 76: DHCP packets showing the UDP transport information and port numbers.

(c) The DHCP Offer contains the IPv4 address being offered to the client along with other network information. In our capture, the server offered 10.0.4.11 to H3S and identified the DHCP server as 10.0.4.21. Figure 77 shows the DHCP Offer information.

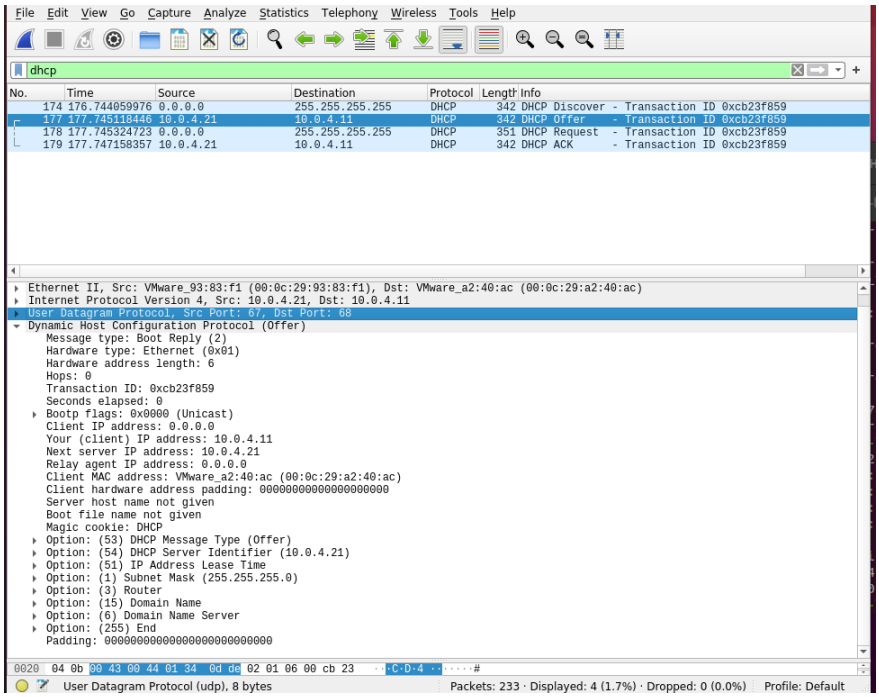


Figure 77: DHCP Offer showing the IPv4 address offered to H3S and the DHCP server information.

Q4: The question is

(a) Open the dhclient.leases file (path /var/lib/dhcp) on H3S and compare its content with that of the subnet (used to provide the IPv4 offer to H3S) as defined in the dhcpd.conf (/etc/dhcp) on M3S. Hint: screenshots of dhclient.leases and dhcpd.conf.

(b) The dhclient.leases has more information than we have added on M3S dhcp server. Find out and state where the additional information comes from.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A4: The answer is

(a) The dhclient.leases file on H3S shows that H3S received the IPv4 address 10.0.4.11. The dhcpd.conf file on M3S shows that the 10.0.4.0/24 subnet has an address range of 10.0.4.10 to 10.0.4.100, so 10.0.4.11 is within the configured range. Figures 78 and 79 show these results.

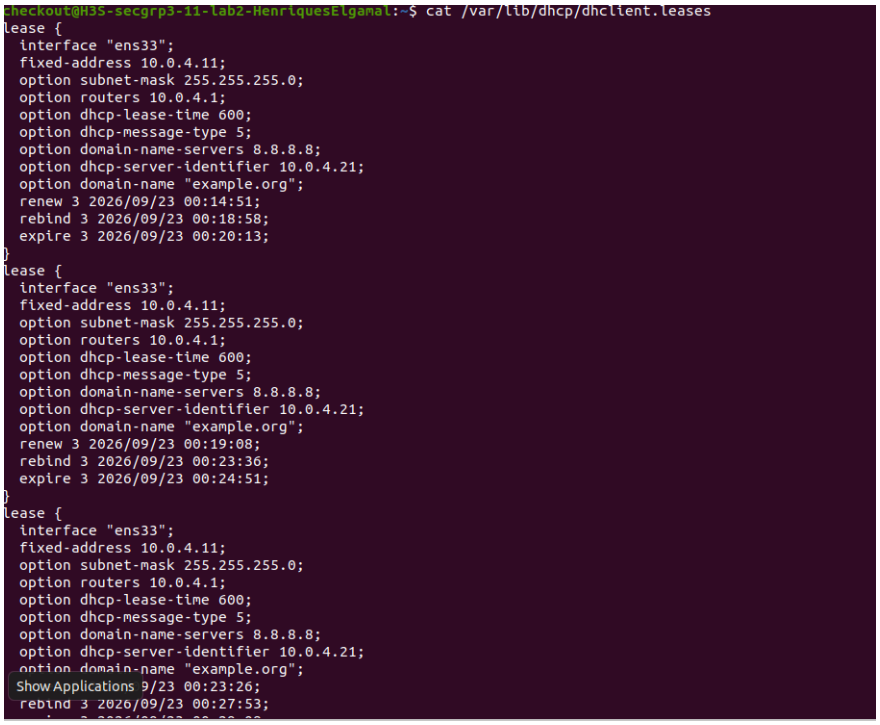


Figure 78: H3S dhclient.leases file showing the IPv4 address and DHCP configuration received by the client.

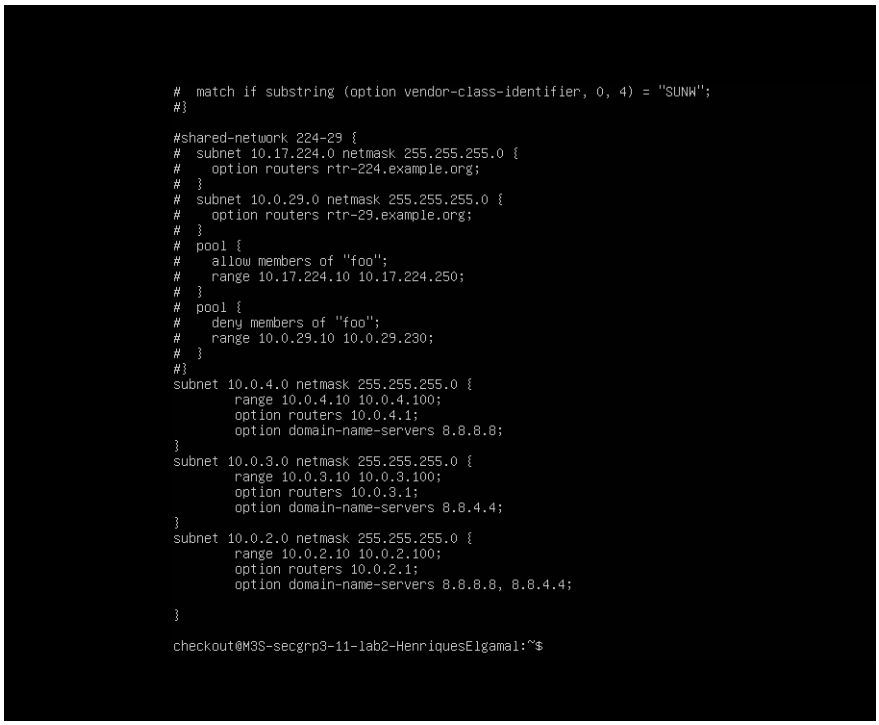


Figure 79: M3S dhcpd.conf file showing the configured DHCP subnet and address range for H3S.

(b) The additional information in dhclient.leases comes from the DHCP server during the DHCP process [10]. The server sends configuration information such as the subnet mask, router, DNS server, lease time, and server identifier to the client.

T4: Done. See the evidence (screenshots in images/L2E1S3/L2E1S3T4 subfolder and attachments, if any, in atts/L2E1S3/L2E1S3T4 subfolder.)

SubA: DHCP PACKET CAPTURING

T5: Done. See the evidence (screenshots in images/L2E1S3/L2E1S3T5 subfolder and attachments, if any, in atts/L2E1S3/L2E1S3T5 subfolder.)

Q5: The question is

(a) Compare the content of dhclient.leases on H3S, H3R and the dhcpd.leases on M3S from the perspective of IPv4 addresses acquired by H3S and H3R and administered by M3S.

(b) Are they consistent with the results of dhcp-lease-list on M3S.

(c) If not, then what would you do to get dhcp-lease-list in sync with the dhclient.leases on H3S and H3R? Try it and show the results. Hint: isc-dhcp-server.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A5: The answer is

(a) The dhclient.leases file on H3S shows that H3S received an address from the 10.0.4.0/24 subnet, while H3R received 10.0.2.83 from the 10.0.2.0/24 subnet. The dhcpd.leases file on M3S also contains the leases for H3S and H3R, showing that M3S administered the addresses. Figures 80, 81, and 82 show these results.

```
checkout@H3R-secgrp3-11-lab2-HenriquesElgamal:~$ cat /var/lib/dhcp/dhclient.leases
lease {
  interface "ens33";
  fixed-address 10.0.2.83;
  option subnet-mask 255.255.255.0;
  option routers 10.0.2.1;
  option dhcp-lease-time 600;
  option dhcp-message-type 5;
  option domain-name-servers 8.8.8.8,8.8.4.4;
  option dhcp-server-identifier 10.0.3.21;
  option domain-name "example.org";
  renew 0 2026/09/27 22:24:49;
  rebind 0 2026/09/27 22:24:49;
  expire 0 2026/09/27 22:24:49;
}
lease {
  interface "ens33";
  fixed-address 10.0.2.83;
  option subnet-mask 255.255.255.0;
  option routers 10.0.2.1;
  option dhcp-lease-time 600;
  option dhcp-message-type 5;
  option domain-name-servers 8.8.8.8,8.8.4.4;
  option dhcp-server-identifier 10.0.3.21;
  option domain-name "example.org";
  renew 0 2026/09/27 22:29:30;
  rebind 0 2026/09/27 22:33:36;
  expire 0 2026/09/27 22:34:51;
}
lease {
```

Figure 80: H3R dhclient.leases showing the IPv4 address received from the DHCP server.

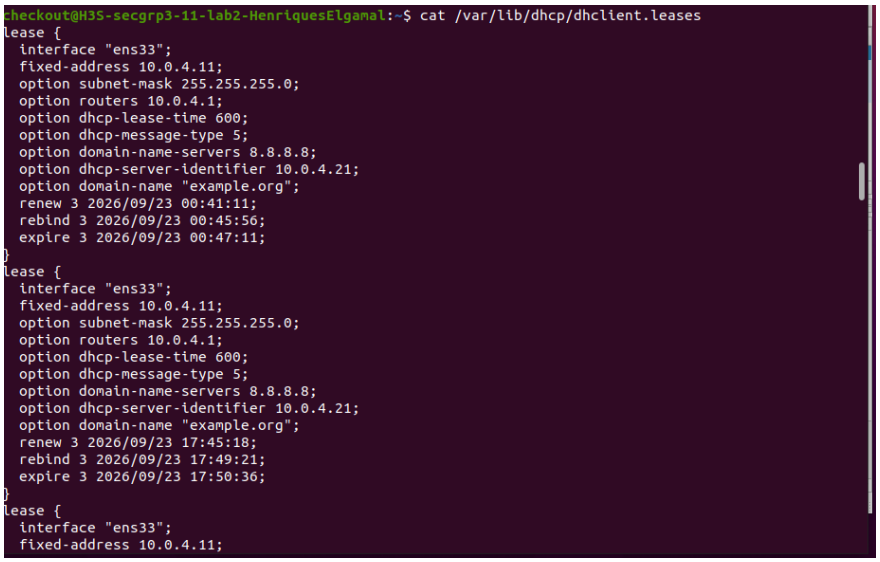


Figure 81: H3S dhclient.leases showing the IPv4 address received from the DHCP server.

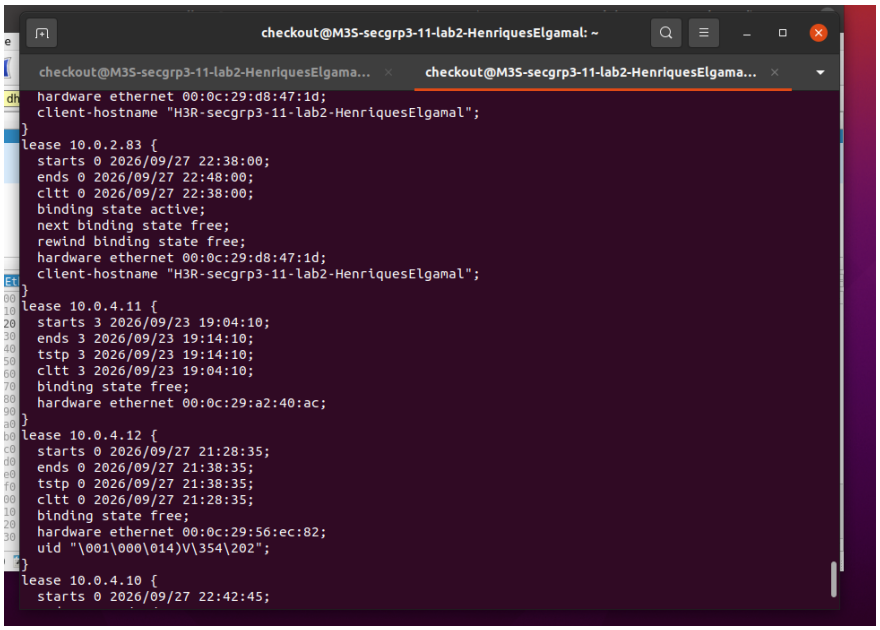


Figure 82: M3S dhcpd.leases showing the DHCP leases administered by the server.

(b) Yes. After the leases were updated, dhcp-lease-list on M3S showed both H3S and H3R and their current IPv4 addresses. Figure 83 shows the final lease list.

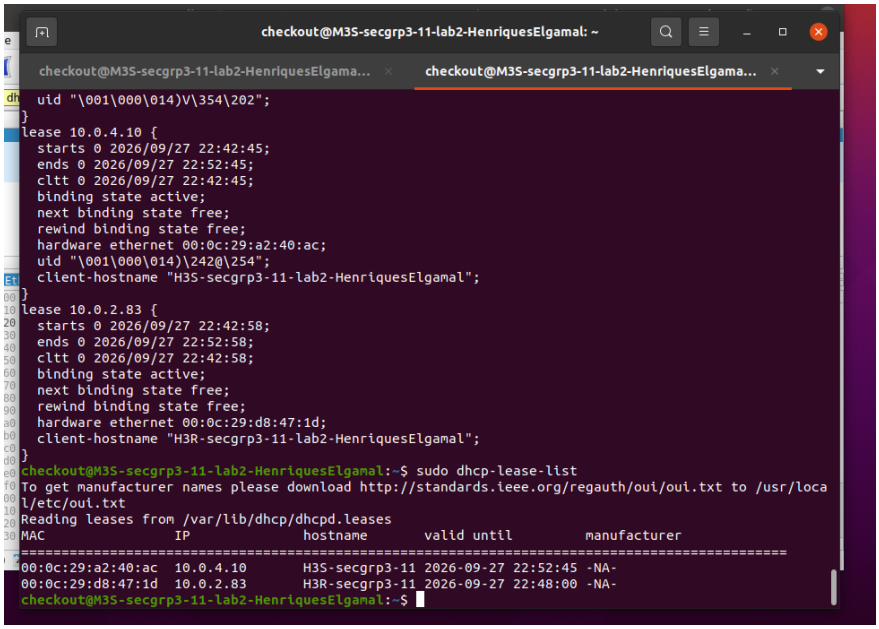


Figure 83: M3S dhcp-lease-list showing the current leases for H3S and H3R.

(c) To get dhcp-lease-list in sync, we restarted the isc-dhcp-server and renewed the DHCP leases on the clients. After doing this, dhcp-lease-list showed the current leases for both H3S and H3R. Figure 84 shows the synchronized results.

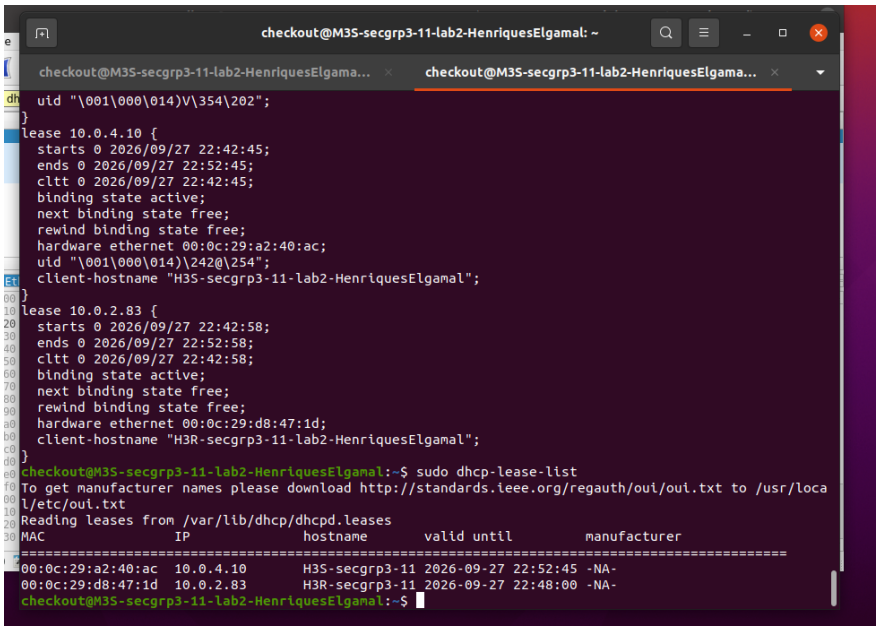


Figure 84: M3S dhcp-lease-list showing the current leases for H3S and H3R.

SubA: DHCP PACKET ANALYSIS

T6: Done. See the evidence (screenshots in images/L2E1S3/L2E1S3T6 subfolder and attachments, if any, in atts/L2E1S3/L2E1S3T6 subfolder.)

Q6: The question is

(a) What are the source and destination IPv4 addresses populated in the DHCP Discover packet on H3R (from H3R to M3R) and on M3S (from M3R to M3S)?

(b) What are the source and destination EUI-48 addresses populated in the same?

(c) Explain the differences between the two sides. Hint: broadcast on 10.0.2.0/24 is converted to unicast on 10.0.3.0/24.

Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A6: The answer is

(a) On H3R, the DHCP Discover packet has a source IPv4 address of 0.0.0.0 and a destination of 255.255.255.255. After M3R relays the packet to M3S, the source is 10.0.3.1 and the destination is 10.0.3.21. Figures 85 and 86 show these addresses

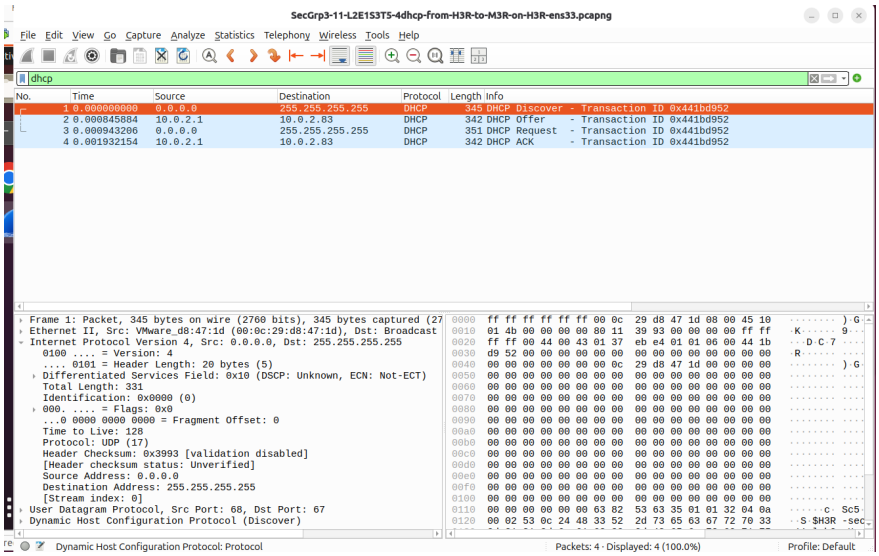


Figure 85: DHCP Discover packet on H3R showing the source and destination IPv4 and EUI-48 addresses.

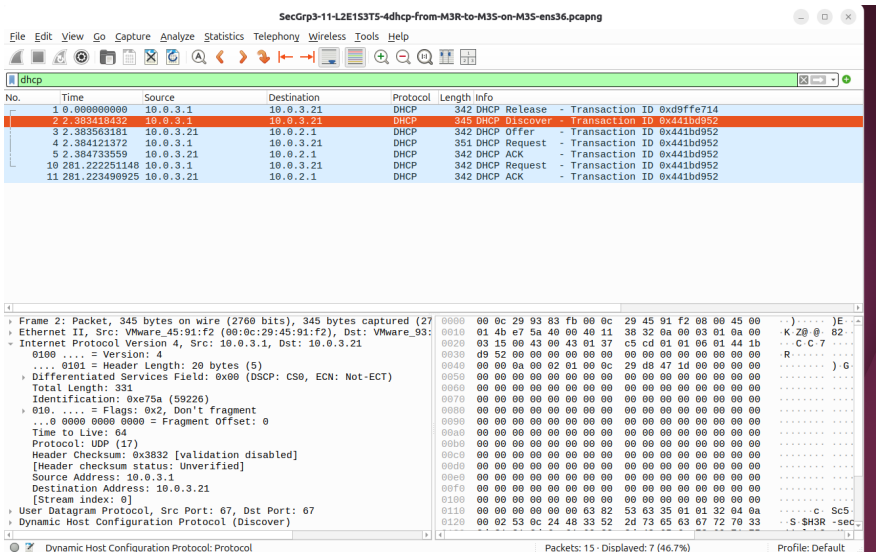


Figure 86: Relayed DHCP Discover packet on M3S showing the source and destination IPv4 and EUI-48 addresses.

(b) On H3R, the source EUI-48 address is 00:0c:29:d8:47:1d and the destination is ff:ff:ff:ff:ff:ff. On M3S, the source EUI-48 address is 00:0c:29:45:91:f2 and the destination is 00:0c:29:93:83:fb. Figures 87

and 88 show these EUI-48 addresses.

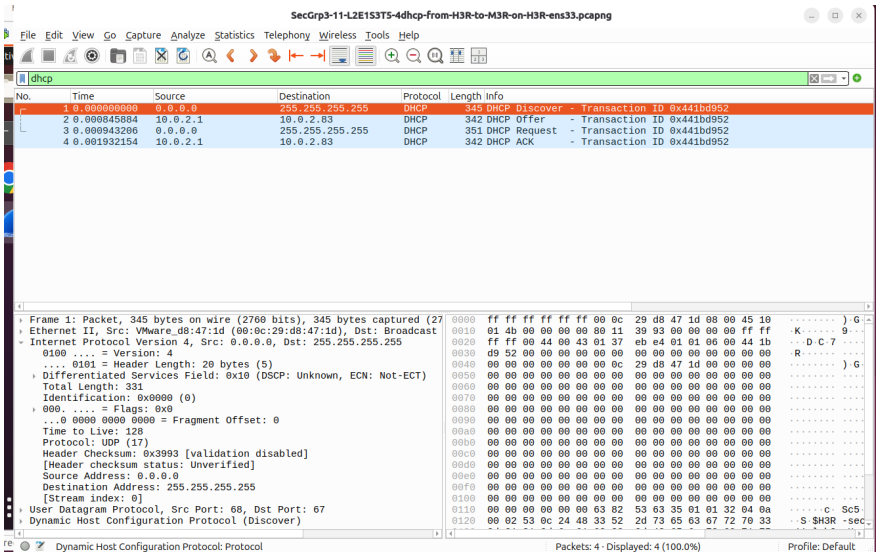


Figure 87: DHCP Discover packet on H3R showing the source and destination IPv4 and EUI-48 addresses.

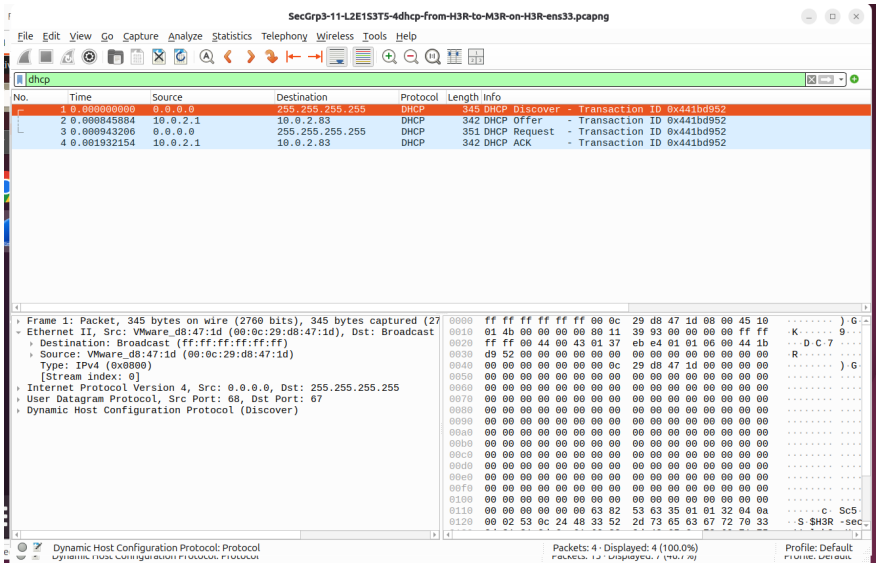


Figure 88: Relayed DHCP Discover on M3S showing the source and destination EUI-48 addresses.

(c) On the H3R side, the DHCP Discover is sent as a broadcast because

H3R does not have an IPv4 address yet. M3R receives this broadcast and relays it as a unicast packet to M3S on the 10.0.3.0/24 subnet. Figures 89 and 90 show the change from broadcast to unicast.

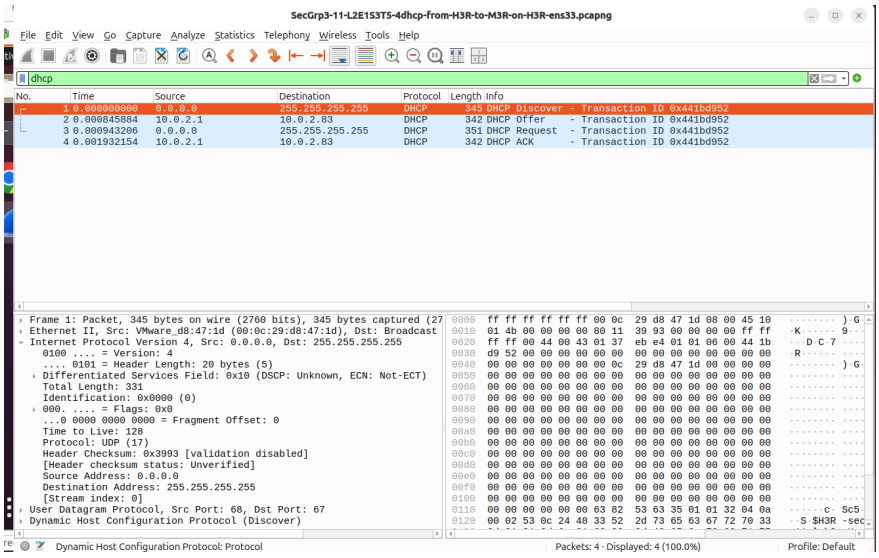


Figure 89: DHCP Discover packet on H3R showing the source and destination IPv4 and EUI-48 addresses.

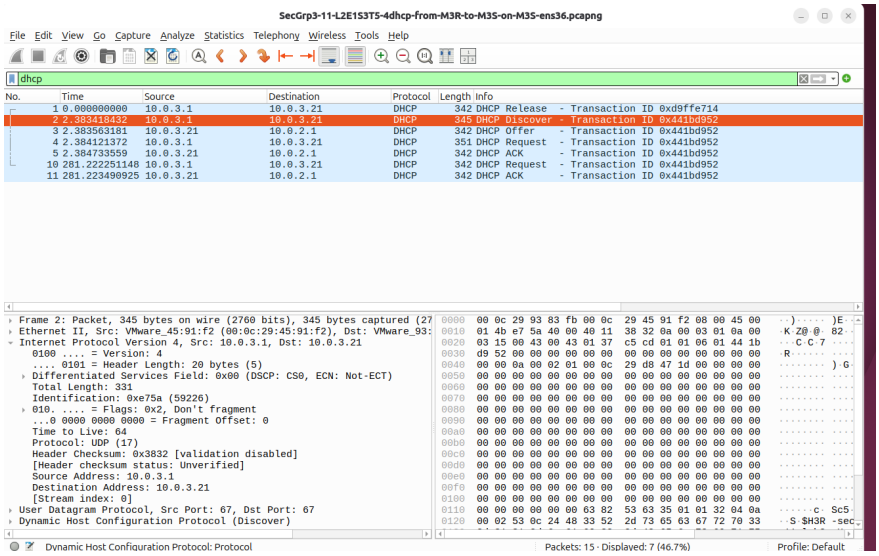


Figure 90: Relayed DHCP Discover packet on M3S showing the source and destination IPv4 and EUI-48 addresses.

Q7: The question is
What are the source and destination transport port numbers populated in the DHCP Discover packet on H3R (from H3R to M3R) and M3S (from M3R to M3S)?
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A7: The answer is
The DHCP Discover packet on H3R uses source port 68 and destination port 67. After M3R relays the Discover packet to M3S, it uses source port 67 and destination port 67. Figures 91 and 92 show the transport port numbers.

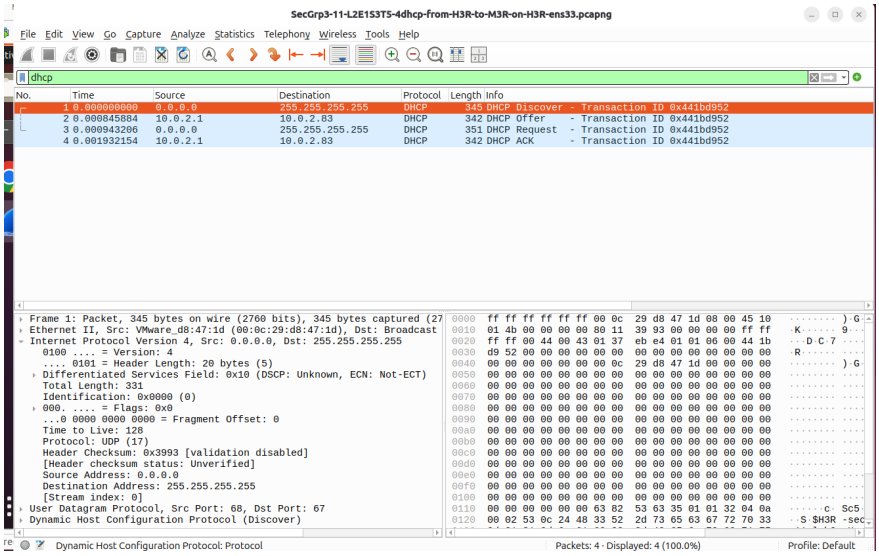


Figure 91: DHCP Discover packet on H3R showing the source and destination IPv4 and EUI-48 addresses.

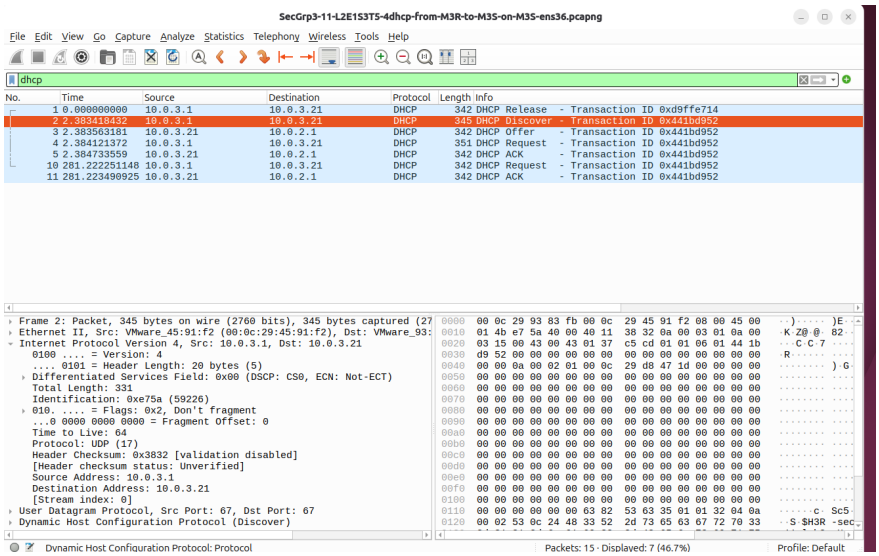


Figure 92: Relayed DHCP Discover packet on M3S showing the source and destination IPv4 and EUI-48 addresses.

Q8: The question is
Using the dhcp packets captured on H3R (from H3R to M3R) and M3S (from M3R to M3S), explain how M3S figures out that the subnet range to be used in response to M3R messages is for a host/client on 10.0.2.0/24 subnet and not on 10.0.3.0/24. Hints: transport port numbers. The payload information of the messages from M3R to M3S.
Provide evidence including AI Apps (e.g., AI Apps (e.g., ChatGPT)) conversations (attachment). Reference the evidence with specific description of the relevant information. Provide citations to external sources.

A8: The answer is
M3S knows that the DHCP request is for a client on the 10.0.2.0/24 subnet because M3R adds relay information to the DHCP message before sending it to M3S. This tells M3S which subnet the original client is on, so M3S selects an address from the 10.0.2.0/24 DHCP range instead of the 10.0.3.0/24 range. Figure 93 shows the DHCP Discover being relayed from M3R to M3S.

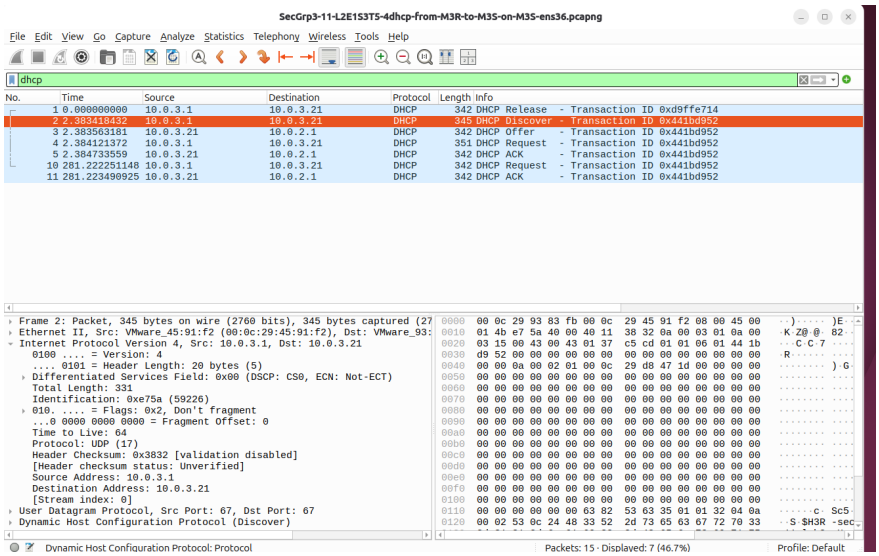


Figure 93: Relayed DHCP Discover packet on M3S showing the source and destination IPv4 and EUI-48 addresses.

4 KLT & EVC

4.1 Key Learning & Takeaways (KLT)

This lab helped us better understand how DHCP, DHCP relay, DNS, and routing work together across different networks. We learned how to configure DHCP for multiple subnets and use Wireshark to see how packets change as they travel between a client, relay, and server.

4.2 Expectancy-Value-Cost (EVC)

Expectancy

After completing this lab, we feel more confident performing the same tasks again. We now have a better understanding of how to configure network interfaces, DHCP servers and relays, and how to use Wireshark to troubleshoot network traffic.

Value

This lab was valuable because it gave us hands-on experience with DHCP, DNS, routing, and packet analysis. These tasks helped us understand how devices receive network information and communicate across different subnets.

Cost

The most time-consuming part of the lab was making sure the correct interfaces, VMnet connections, and IPv4 addresses were configured on each virtual machine. Troubleshooting DHCP and relay problems also took extra time because one incorrect network setting could prevent the client from receiving an address. The effort was worth it because troubleshooting these problems helped us better understand how the network was working.

5 Observations, Suggestions & Best Practices

5.1 Lab

5.1.1 Observations

During the lab, we learned how DHCP works across multiple subnets using a DHCP server and relay. One thing that surprised us was how the DHCP Discover packet changed from a broadcast on the client network to a unicast packet after being forwarded by the relay. We also got stuck when some of the interface names in our VMs were different from the ones shown in the lab instructions, so we had to verify the correct interfaces before continuing.

5.1.2 Suggestions

The lab instructions could be clearer about how to identify the correct network interface when the interface names are different from the examples. It would also help to include a small network diagram for each major step showing which VMnet, interface, and IPv4 address should be connected to each machine. This would make it easier to check the configuration before moving to the next task.

5.1.3 Best Practices

A good practice during this lab was checking the IP addresses and interfaces after every major network change instead of assuming the configuration was correct. We also checked that services such as the DHCP server and DHCP relay were running before troubleshooting other problems. If we repeated the lab, we would verify all VMnet connections, interface names, and IP addresses at the beginning to avoid configuration problems later.

6 Lab References

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7 Acknowledgments

We would like to acknowledge ChatGPT for helping us throughout this lab. We used it to better understand networking concepts, troubleshoot problems, and clarify parts of the lab when we needed additional help. It was especially helpful when we needed another explanation of a concept or guidance on how to approach an issue. We would also like to acknowledge TA Korbinian as well for helping us when we needed it such as troubleshooting problems and understanding what certain rubric standards meant.